The niche of Romik's ambidextrous sofa in convex-linear data, and a stability bound under a curvature condition

Vico Bonfioli
vicobonfioli@gmail.com

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Abstract

Baek proved that Gerver's sofa is the maximum-area moving sofa for a corridor with one corner, by writing a sofa as a convex cap minus a niche and obtaining optimality from the concavity of a quadratic upper bound. We carry that architecture to the ambidextrous moving sofa problem, which is open, and to Romik's candidate Σ of area $A_R^* = 1.6449552184\ldots$

The niche of an ambidextrous sofa is expressed exactly in convex-linear data: writing $H(\theta) = h_K(\mu_\theta)$ for the cap's support function, the two arms α_1, α_2 and the reach σ are affine in H, and $|N| = \int_0^{\pi/2}[\frac{1}{2}(\alpha_2^+)^2 + \frac{1}{2}(\sigma-\alpha_1)^2 - \frac{1}{2}(\alpha_1^-)^2]\,dt$ whenever the associated sweep is injective. Injectivity follows from a single linear condition (RC), that the cap's radius of curvature be at most the corridor width; the proof reduces each of the three required statements to a forced harmonic oscillator whose kernel is non-negative because the rotation angle is $\pi/2$. (RC) is classical — by the dual of Blaschke's rolling theorem it says the cap slides freely inside a unit ball, up to the corridor facets — so what is new is its use.

For the functional $Q = |C_2| - 2V$ we prove $Q(\Sigma) = A_R^*$; that $\delta Q(\Sigma) = 0$, so Romik's ODE families are exactly the critical-point equations of Q, identically in his constants; and that $\delta^2 Q < 0$, with an explicit constant. Two Cauchy–Schwarz steps decouple the two halves of $[0, \pi]$ and reduce the second variation to a pair of Sturm–Liouville eigenvalue problems, certified from below by Sturm oscillation; this gives $\frac{1}{2}\delta^2 Q \le -\frac{2}{3}\|\eta\|^2_{L^2(0,\pi)}$. Not splitting at all turns the second variation into a single 2×2 system whose first eigenvalue is the sharp constant $c^* = 0.7309566\ldots$, certified to exceed $\frac{73}{100}$. On the convex domain $\mathcal{D}$ cut out by (RC), a separation inequality and Σ's own sign pattern,

$$|T| \;\le\; A_R^* - \tfrac{73}{100}\,\|H - H_\Sigma\|^2_{L^2(0,\pi)}$$

for every ambidextrous sofa T with cap data in $\mathcal{D}$, and Σ is the unique member attaining A_R^*. The domain is a C^2-neighbourhood of Σ: its radius along $\sin(2k\theta)$ scales like k^{-2}.

Two limits are recorded rather than smoothed over. The separation inequality is independent of (RC) and cannot be removed. And V is the exact flux of the advancing wedge boundary, so $V \ge |N|$ by Reynolds' transport theorem and Q bounds nothing where the sweep fails to be injective; in particular the construction does not recover Gerver's sofa, whose cap violates (RC). Optimality of Σ remains open.

1 Setting

Write L for the closed L-shaped hallway of unit width with its corner at the origin, $\mu_t = (\cos t, \sin t)$, $\nu_t = (-\sin t, \cos t)$, and for a rotation path $c : [0, \pi/2] \to \mathbb{R}^2$ let

$$C_t = \{p : \langle p - c(t), \mu_t\rangle \le 1\} \cap \{p : \langle p - c(t), \nu_t\rangle \le 1\}, \qquad Q_t = \{p : \langle p - c(t), \mu_t\rangle < 0\} \cap \{p : \langle p - c(t), \nu_t\rangle < 0\},$$

so that the hallway in position t is $H_t = C_t \setminus Q_t$. The one-sided sofa is $S = \bigcap_t H_t$, and writing $C = \bigcap_t C_t$ and $U = \bigcup_t Q_t$ we have $S = C \setminus U$ with C convex.

Let $\rho(x, y) = (x, 1 - y)$ be the reflection in the line $y = \frac{1}{2}$. Romik's ambidextrous sofa is

$$\Sigma = S \cap \rho(S) = C_2 \setminus \bigl(U \cup \rho U\bigr), \qquad C_2 := C \cap \rho C \text{ convex}, \tag{1}$$

with $|\Sigma| = A_R^* = 1.6449552184\ldots$, and c is Romik's piecewise path with breakpoints at β and $\pi/2 - \beta$, where

$$\beta = \arctan\Bigl(\tfrac{1}{2}\bigl(\sqrt[3]{\sqrt{2}+1} - \sqrt[3]{\sqrt{2}-1}\bigr)\Bigr) = 0.2896538208\ldots \tag{2}$$

Baek's argument for the one-corner problem builds an overestimating region whose area is a quadratic functional on a convex domain of convex bodies, and proves it concave by exhibiting it as a linear functional minus Mamikon regions, whose areas are convex quadratics. Concavity reduces global optimality to a first-order condition. The construction is organised around $S = K \setminus \mathcal{N}(K)$: one cap, one niche.

By (1) the ambidextrous analogue has *two* niches, and

$$|U \cup \rho U| = |U| + |\rho U| - |U \cap \rho U|. \tag{3}$$

The last term appears in the upper bound with a $+$ sign. Since concavity is obtained by *subtracting* convex quadratics, an added overlap term is not covered by the mechanism. The purpose of this note is Theorem 8: the term is zero.

2 The niches are disjoint

Lemma 1 (Niche ceiling)**.** *For every $t \in [0, \pi/2]$ we have $Q_t \subseteq \{(x, y) : y \le c_y(t)\}$.*

Proof. Q_t is the open convex cone with apex $c(t)$ spanned by $-\mu_t$ and $-\nu_t$, so $q \in Q_t$ means $q - c(t) = -a\mu_t - b\nu_t$ with $a, b > 0$. For $t \in [0, \pi/2]$ both $\sin t \ge 0$ and $\cos t \ge 0$, hence

$$q_y - c_y(t) = -a \sin t - b \cos t \le 0. \qquad \square$$

Corollary 2. *Put $M := \max_{t \in [0,\pi/2]} c_y(t)$. Then $U \subseteq \{y \le M\}$ and $\rho U \subseteq \{y \ge 1 - M\}$. If $M < \frac{1}{2}$ then $U \cap \rho U = \emptyset$.*

Proof. The first inclusion is Lemma 1 applied to each t; the second is its image under ρ. If $M < \frac{1}{2}$ then $1 - M > \frac{1}{2} > M$, so the two half-planes are disjoint. $\square$

It remains to evaluate M for Romik's path. On the middle piece $[\beta, \pi/2 - \beta]$ Romik's path is

$$c(t) = R(t)v(t) + \kappa, \qquad v(t) = \begin{pmatrix} f_1 \cos\frac{t}{2} + f_2 \sin\frac{t}{2} - 1 \\ -f_2 \cos\frac{t}{2} + f_1 \sin\frac{t}{2} - 1 \end{pmatrix}, \tag{4}$$

with $R(t)$ the rotation by t, and the two facts we use are

$$\kappa_y = \tfrac{1}{2} \quad \text{exactly}, \qquad f_2 = (1 - \sqrt{2})\, f_1. \tag{5}$$

Proposition 3. *For $t \in [\beta, \pi/2 - \beta]$,*

$$c_y(t) - \tfrac{1}{2} = f_1 \sin \tfrac{3t}{2} - f_2 \cos \tfrac{3t}{2} - (\sin t + \cos t) = f_1 \sqrt{4 - 2\sqrt{2}} \sin\Big(\tfrac{3t}{2} + \tfrac{\pi}{8}\Big) - \sqrt{2} \sin\Big(t + \tfrac{\pi}{4}\Big). \quad (6)$$

In particular c_y is symmetric about $t = \pi/4$ on this interval, and

$$M = c_y(\pi/4) = \tfrac{1}{2} - \Big(\sqrt{2} - f_1 \sqrt{4 - 2\sqrt{2}}\Big). \quad (7)$$

Proof. By (4) and $\kappa_y = \frac{1}{2}$, $c_y(t) - \frac{1}{2} = \sin t\, v_x + \cos t\, v_y$. Write $S = \sin \frac{t}{2}$, $C = \cos \frac{t}{2}$, so $\sin t = 2SC$ and $\cos t = C^2 - S^2$. Expanding and collecting the coefficients of f_1 and f_2 gives

$$\sin t\, v_x + \cos t\, v_y = f_1 S(3C^2 - S^2) + f_2 C(3S^2 - C^2) - (\sin t + \cos t),$$

and the triple-angle identities $S(3C^2 - S^2) = \sin \frac{3t}{2}$, $C(3S^2 - C^2) = -\cos \frac{3t}{2}$ give the first equality in (6). For the second, substitute $f_2 = (1 - \sqrt{2}) f_1$ from (5): the bracket becomes $\sin \frac{3t}{2} + (\sqrt{2} - 1) \cos \frac{3t}{2}$, of amplitude $\sqrt{1 + (\sqrt{2} - 1)^2} = \sqrt{4 - 2\sqrt{2}}$ and phase $\arctan(\sqrt{2} - 1) = \pi/8$.

Both sinusoids in (6) are invariant under $t \mapsto \pi/2 - t$: indeed $\sin(\frac{3}{2}(\pi/2 - t) + \pi/8) = \sin(7\pi/8 - \frac{3t}{2}) = \sin(\pi/8 + \frac{3t}{2})$ and $\sin(3\pi/4 - t) = \sin(\pi/4 + t)$. Hence c_y is symmetric about $\pi/4$, where both sinusoids attain the value 1 simultaneously; (7) follows. □

That $\pi/4$ maximises c_y over all of $[0, \pi/2]$, and not merely over the middle piece, follows from the corresponding closed form on the outer pieces.

Proposition 4. *For $t \in [0, \beta]$,*

$$c_y(t) = \tfrac{1}{2} + a_1 \sin 2t - \sin t - \tfrac{1}{2} \cos t, \quad (8)$$

and c_y is strictly increasing there, so $\max_{[0,\beta]} c_y = c_y(\beta)$. By the symmetry $c_y(t) = c_y(\pi/2 - t)$ the same bound holds on $[\pi/2 - \beta, \pi/2]$. Consequently $M = c_y(\pi/4)$ as claimed in (7).

Proof. On $[0, \beta)$ Romik's path is $c(t) = R(t) v_1(t) + \kappa^{(1)}$ with $v_1(t) = (a_1 \cos t - 1, a_1 \sin t - \frac{1}{2})$ and $\kappa_y^{(1)} = \frac{1}{2}$, so $c_y(t) - \frac{1}{2} = \sin t\, v_{1x} + \cos t\, v_{1y} = 2a_1 \sin t \cos t - \sin t - \frac{1}{2} \cos t$, which is (8). Differentiating,

$$c_y'(t) = 2a_1 \cos 2t - \cos t + \tfrac{1}{2} \sin t \;\geq\; 2a_1 \cos 2\beta - 1 \;=\; 0.4650\ldots > 0 \qquad (t \in [0, \beta]),$$

using $\cos 2t \geq \cos 2\beta$ and $\cos t - \frac{1}{2} \sin t \leq 1$ for $t \geq 0$. Hence c_y is strictly increasing on $[0, \beta]$, with $c_y(\beta) = 0.2143795161\ldots$, which is below $c_y(\pi/4) = 0.3878381292\ldots$. □

Remark 5. Evaluating (8) and (6) at $t = \beta$ gives the same value to $4.4 \cdot 10^{-31}$ at 30-digit precision. Since the two expressions were derived from different pieces of Romik's path, this is an independent check on both, and on (10).

3 A closed form for f_1

Romik's constant f_1 is recorded in the literature as the decimal 1.202938908156911389. The following determines it in closed form. Let

$$a_1 = \tfrac{1}{4}\sqrt{4 + \sqrt[3]{71+8\sqrt{2}} + \sqrt[3]{71-8\sqrt{2}}} = 0.8752873624\ldots \tag{9}$$

Proposition 6.

$$f_1 = \frac{\frac{4}{3}\, a_1 \cos\beta}{\cos\frac{\beta}{2} + (1-\sqrt{2})\sin\frac{\beta}{2}}. \tag{10}$$

Proof. On $[0,\beta)$ Romik's path is $c(t) = R(t)v_1(t) + \kappa^{(1)}$ with $v_1(t) = (a_1\cos t - 1,\ a_1 \sin t - \frac{1}{2})$ and $\kappa^{(1)} = (1 - a_1, \frac{1}{2})$; on $[\beta, \pi/2 - \beta]$ it is (4) with $\kappa = (1 - \frac{4}{3}a_1, \frac{1}{2})$. Both κ's have second coordinate $\frac{1}{2}$ and their first coordinates differ by exactly $\frac{1}{3}a_1$. Continuity at $t = \beta$ therefore gives $R(\beta)\big(v_1(\beta) - v(\beta)\big) = (-\frac{1}{3}a_1, 0)$, so

$$v_1(\beta) - v(\beta) = R(-\beta)\big(-\tfrac{1}{3}a_1, 0\big) = \big(-\tfrac{1}{3}a_1\cos\beta,\ \tfrac{1}{3}a_1\sin\beta\big).$$

Reading the first coordinate and using $f_2 = (1-\sqrt{2})f_1$,

$$\big(a_1\cos\beta - 1\big) - \Big(f_1\big(\cos\tfrac{\beta}{2} + (1-\sqrt{2})\sin\tfrac{\beta}{2}\big) - 1\Big) = -\tfrac{1}{3}a_1\cos\beta,$$

which rearranges to (10). □

Remark 7. Evaluating (10) at 50 digits gives $f_1 = 1.2029389081569113890702 23\ldots$, so the tabulated decimal is its rounding. As an independent check, the *second* coordinate of the junction relation, which was not used in the derivation, is satisfied to $1.7 \cdot 10^{-51}$ at the same precision.

4 Main theorem

Theorem 8. *For Romik's ambidextrous sofa, $U \cap \rho U = \emptyset$, and consequently*

$$|\Sigma| = |C_2| - |U| - |\rho U| = |C_2| - 2|U|.$$

Proof. By Proposition 3, $M < \frac{1}{2}$ is equivalent to $f_1\sqrt{4-2\sqrt{2}} < \sqrt{2}$, i.e. to

$$f_1^2 < \frac{2+\sqrt{2}}{2}. \tag{11}$$

With f_1 given by (10) in terms of the algebraic numbers (2) and (9), both sides of (11) are explicit algebraic numbers, and

$$f_1^2 = 1.4470620167577420940\ldots, \qquad \frac{2+\sqrt{2}}{2} = 1.7071067811865475244\ldots,$$

so (11) holds with margin $0.2600447644\ldots$. Corollary 2 gives $U \cap \rho U = \emptyset$, and then (3) together with (1). Finally $|\rho U| = |U|$ because ρ is an isometry. □

Corollary 9. $M = 0.3878381292441942964\ldots$, *the two niches are separated by the horizontal band of width $1 - 2M = 0.2243237415116114\ldots$ about $y = \frac{1}{2}$, and the ambidextrous upper bound requires only* one *niche to be modelled: one core and two tails, the same shape as in the one-corner problem.*

5 The quotient reduction and a conditional statement

Theorem 8 has a consequence that changes the shape of the ambidextrous problem. Since $U \subseteq \{y \le M\}$ with $M < \frac{1}{2}$, the whole lower niche lies below the symmetry axis, so intersecting with the lower half-strip deletes ρU outright.

Corollary 10 (Quotient reduction). *Put $K^- := C_2 \cap \{y \le \frac{1}{2}\}$, a convex body. Then*

$$\Sigma \cap \{y \le \tfrac{1}{2}\} = K^- \setminus U, \qquad |\Sigma| = 2\big(|K^-| - |U \cap K^-|\big).$$

That is a convex cap minus *one* niche: the shape Baek's argument is built on. Numerically, at $n = 1441$ hallways, $|\Sigma| = 1.645059395$ and $|\Sigma \cap \{y \le \frac{1}{2}\}| = 0.822529698$, whose double reproduces $|\Sigma|$ to $3.8 \cdot 10^{-15}$.

Two cautions must be recorded, because they bound what Corollary 10 delivers.

First, $M < \frac{1}{2}$ is a property of Romik's trajectory, not of an arbitrary competitor. A competitor with $M \ge \frac{1}{2}$ has overlapping niches, and by (3) overlap *increases* $|\Sigma|$; so it cannot be assumed away on extremal grounds.

Second — and this is what makes the reduction usable — connectedness does appear to exclude it, but not for the reason one first tries. A *single* pair $Q_t \cup \rho Q_t$ removes a full vertical segment, and hence disconnects any body meeting both sides of it, exactly when the apex height exceeds a threshold $h^*(t)$; and computation gives $h^*(t) > \frac{1}{2}$ throughout $(0, \pi/2)$, with $h^*(t) \to \frac{1}{2}$ as $t \to 0$, $h^*(\pi/4) = 0.5100$ and $h^*(1.52) = 0.6967$. So no single-t argument forces $M \le \frac{1}{2}$; all it gives is the pointwise constraint $c_y(t) \le h^*(t)$.

In fact a single pair does force it, and the earlier reading to the contrary was an artifact of testing the vertical line at a fixed distance from the apex.

Theorem 11 (Connectedness ceiling). *Let $p = c(t_0)$ with $t_0 \in (0, \pi/2)$ and suppose $p_y > \frac{1}{2}$. Put*

$$\varepsilon_0 := \frac{2p_y - 1}{2 \tan t_0} > 0.$$

Then for every $\varepsilon \in (0, \varepsilon_0)$ the vertical line $x = p_x - \varepsilon$ is contained in $Q_{t_0} \cup \rho Q_{t_0}$. Hence any ambidextrous moving sofa omits the open vertical strip $\{p_x - \varepsilon_0 < x < p_x\}$, and a connected one lies entirely in one of the two half-planes it separates.

Proof. Fix $\varepsilon > 0$ and consider $q = (p_x - \varepsilon, y)$. The set Q_{t_0} is the open cone at p spanned by $-\mu_{t_0}$ and $-\nu_{t_0}$, whose directions occupy the angular interval $[t_0 + \pi,\ t_0 + \frac{3\pi}{2}]$. Writing $h := p_y - y$, the direction $q - p = (-\varepsilon, -h)$ has angle $\pi + \arctan(h/\varepsilon)$ when $h > 0$, so $q \in Q_{t_0}$ iff $\arctan(h/\varepsilon) \ge t_0$, that is iff

$$y \le p_y - \varepsilon \tan t_0.$$

Applying ρ and using $\rho q - \rho p = R(q - p)$ with $R = \mathrm{diag}(1, -1)$, the reflected cone ρQ_{t_0} has apex $(p_x, 1 - p_y)$ and occupies the angular interval $[\frac{\pi}{2} - t_0,\ \pi - t_0]$, giving by the same computation

$$q \in \rho Q_{t_0} \iff y \ge 1 - p_y + \varepsilon \tan t_0.$$

The two sets therefore cover the whole line exactly when $p_y - \varepsilon \tan t_0 \ge 1 - p_y + \varepsilon \tan t_0$, i.e. when $2\varepsilon \tan t_0 \le 2p_y - 1$, i.e. when $\varepsilon \le \varepsilon_0$. Since $p_y > \frac{1}{2}$ we have $\varepsilon_0 > 0$ and the range is nonempty. Taking the union over $\varepsilon \in (0, \varepsilon_0)$ gives the strip. $\square$

Remark 12. The criterion is sharp in both directions: if $p_y \le \frac{1}{2}$ then $\varepsilon_0 \le 0$ and the two cones leave a gap at *every* distance from the apex, which is why $\frac{1}{2}$ is exactly the threshold. It also explains a numerical trap: ε_0 can be very small — at $t_0 = 1.3$, $p_y = 0.55$ it is 0.0139 — so sampling the line at a fixed distance such as 0.02 detects no covering and suggests, wrongly, that a single pair never separates. The criterion was checked against direct computation at seven pairs (t_0, p_y) on both sides of ε_0, and at three pairs with $p_y < \frac{1}{2}$.

Granting that an ambidextrous sofa cannot lie in one of the two half-planes separated by such a strip — it must meet both arms of the hallway H_{t_0} whose inner corner is p — Theorem 11 yields $\max_t c_y(t) \le \frac{1}{2}$ for every connected ambidextrous moving sofa, and the hypothesis of the next proposition is automatic.

Proposition 13 (Conditional form)**.** *Let S be an ambidextrous moving sofa with rotation path c satisfying $\max_t c_y(t) \le \frac{1}{2}$. Then its two niches are disjoint and $|S| = 2(|K^-| - |U \cap K^-|)$ with K^- convex and U a single niche. In particular the area functional of the ambidextrous problem restricted to this class is of the cap-minus-one-niche form to which the concavity method applies. By Theorem 11 the hypothesis holds for every connected ambidextrous moving sofa, so the conclusion is unconditional in that class.*

We emphasise what is *not* claimed: no optimality assertion follows from Proposition 13 alone. Baek's route additionally requires an existence and regularity step (a maximum monotone sofa of rotation angle $\pi/2$ satisfying an injectivity condition), and his balancing argument for that step does not transfer, for a reason worth recording. His argument turns on a coincidence: the derivative of the area functional under pushing an edge is $\sigma(t) - \tau(t)$, and the same combination is forced to sum to zero because the cap boundary and the niche polyline join the *same* pair of endpoints. In the quotient, ∂K^- acquires the horizontal cut at $y = \frac{1}{2}$, where the polyline contributes nothing, and the identity becomes $\sum_{t \ne \mathrm{cut}} (\sigma - \tau)(v_t \cdot u_0) = \sigma_{\mathrm{cut}} > 0$, which is consistent with $\sigma \le \tau$ throughout. One obtains an inequality where Baek obtains an equality, and it is the equality that his argument consumes.

6 A relation between Romik's constants

The phase junctions of Σ admit a characterisation that ties two of Romik's constants together. Recall a_1 from (9) and β from (2).

Theorem 14. $4a_1 \sin\beta = 1$.

Proof. Write $u := \sqrt[3]{\sqrt{2}+1} - \sqrt[3]{\sqrt{2}-1}$, so that $\tan\beta = u/2$ by (2) and hence $\sin\beta = u/\sqrt{4+u^2}$, and $w := \sqrt[3]{71+8\sqrt{2}} + \sqrt[3]{71-8\sqrt{2}}$, so that $a_1 = \frac{1}{4}\sqrt{4+w}$ by (9). Then

$$4a_1 \sin\beta = \frac{u\sqrt{4+w}}{\sqrt{4+u^2}},$$

so the assertion is equivalent to $u^2(4+w) = 4 + u^2$, that is to

$$u^2\,(3+w) = 4. \tag{12}$$

Put $p := \sqrt[3]{\sqrt{2}+1}$ and $q := \sqrt[3]{\sqrt{2}-1}$. Then $pq = 1$ and $p^3 - q^3 = 2$, so

$$u^3 = (p-q)^3 = p^3 - q^3 - 3pq(p-q) = 2 - 3u,$$

i.e.

$$u^3 + 3u = 2. \tag{13}$$

Set $x := u^2$. Then (13) reads $u(x+3) = 2$; squaring and using $u^2 = x$ gives

$$x(x+3)^2 = 4, \qquad \text{i.e.} \qquad x^3 + 6x^2 + 9x - 4 = 0. \tag{14}$$

Now let $W := 4/x - 3$. Clearing denominators,

$$W^3 - 51W - 142 = \frac{(4-3x)^3 - 51(4-3x)x^2 - 142x^3}{x^3} = \frac{-16\bigl(x^3 + 6x^2 + 9x - 4\bigr)}{x^3} = 0$$

by (14), the middle equality being the polynomial identity $(4-3x)^3 - 51(4-3x)x^2 - 142x^3 = -16(x^3 + 6x^2 + 9x - 4)$.

On the other hand w satisfies the same cubic: with $A := \sqrt[3]{71 + 8\sqrt{2}}$, $B := \sqrt[3]{71 - 8\sqrt{2}}$ one has $AB = \sqrt[3]{71^2 - 128} = \sqrt[3]{4913} = 17$ and $A^3 + B^3 = 142$, so $w^3 = A^3 + B^3 + 3AB\,w = 142 + 51w$. The cubic $W^3 - 51W - 142$ has discriminant $-4(-51)^3 - 27(-142)^2 = 530604 - 544428 = -13824 < 0$ and therefore exactly one real root; both $4/x - 3$ and w are real roots, so $4/x - 3 = w$. This is (12). $\square$

Corollary 15. *$2\tan\beta$ is the real root of $u^3 + 3u = 2$ — equivalently $4\tan^3\beta + 3\tan\beta = 1$, the cubic already recorded in the literature — and*

$$\sqrt[3]{71 + 8\sqrt{2}} + \sqrt[3]{71 - 8\sqrt{2}} = u^4 + 6u^2 + 6.$$

Proof. The first assertion is (13). For the second, (14) gives $(x+3)^2 = 4/x$, so $w = 4/x - 3 = (x+3)^2 - 3 = x^2 + 6x + 6$ with $x = u^2$. $\square$

Proof of the junction statement. On $[0, \beta)$ Romik's path is $c(t) = R_t v(t) + \kappa^{(1)}$ with $v(t) = (a_1 \cos t - 1,\ a_1 \sin t - \frac{1}{2})$. Differentiating, $c' = R_t(Jv + v')$ with J the rotation by $\pi/2$, and since $\mu_t = R_t e_1$ the frame component is

$$c'(t) \cdot \mu_t = (Jv + v')_x = -v_y - a_1 \sin t = -\bigl(a_1 \sin t - \tfrac{1}{2}\bigr) - a_1 \sin t = \tfrac{1}{2} - 2a_1 \sin t.$$

This vanishes precisely when $\sin t = 1/(4a_1)$. On the other hand the phase boundary of Σ at $t = \beta$ is where the SOL1 phase ends, and by Proposition 4 the two closed forms for c_y agree there; the same computation on the SOL5 phase gives $c' \cdot \nu_t = 2a_1 \cos t - \frac{1}{2}$, vanishing when $\cos t = 1/(4a_1)$, i.e. at $t = \pi/2 - \beta$. Since the two junctions are β and $\pi/2 - \beta$ and $\cos(\pi/2 - \beta) = \sin\beta$, both conditions reduce to $\sin\beta = 1/(4a_1)$. $\square$

Remark 16 (Attribution)**.** Neither (13) nor (14) is new. The standard description of Σ gives its area as $x + \arctan y$ where $x^2(x+3) = 8$ and $y(4y^2 + 3) = 1$; the second of these is exactly (13) with $y = \tan\beta = u/2$, and the first is (14) shifted, since $x = u^2 + 1$ turns $x^3 + 3x^2 - 8$ into $u^6 + 6u^4 + 9u^2 - 4$. Equivalently

$$A_R^* = u^2 + 1 + \arctan(u/2), \qquad u^3 + 3u = 2,$$

so a single algebraic number generates both parts of the area. What we have not found stated is the relation $4a_1 \sin\beta = 1$ of Theorem 14 between the shape constant a_1 and β, nor $w = u^4 + 6u^2 + 6$; but both are elementary consequences of the two standard cubics, and we make no priority claim. Every step was checked numerically at 60 digits, and the two polynomial steps are machine-checked (`u_sq_cubic`, `cardano_substitution`).

Theorem 14 has a consequence for the transfer of Baek's method. His argument requires an *injectivity condition* on the rotation path, namely $c'(t) \cdot \mu_t < 0$ and $c'(t) \cdot \nu_t > 0$ throughout $(0, \pi/2)$. By the computation above, for Σ these quantities vanish exactly at β and $\pi/2 - \beta$ and have the wrong sign outside $[\beta, \pi/2 - \beta]$: at $t = 0.05$ one finds $c' \cdot \mu_t = +0.4125$. So Σ satisfies the injectivity condition on its middle phase and violates it on both outer phases, and the failure boundary is exactly the phase junction.

Remark 17 (Where the difficulty of the ambidextrous problem sits)**.** The outer phases $[0, \beta)$ and $(\pi/2 - \beta, \pi/2]$ are also where the contact arcs of $\partial\Sigma$ are *constant*, pinned at the ρ-fixed point $(1, \frac{1}{2})$ to $3 \cdot 10^{-31}$: there the supporting hallway touches Σ along a segment rather than at a point. Three separate obstructions therefore coincide on those phases — the contact structure degenerates, second variation arguments lose the contact point, and the injectivity hypothesis of the one-corner proof fails.

7 The ambidextrous problem as a single family of angle range π

Finally we record a reformulation which explains why the transfer is delicate. Since ρ carries the hallway at angle s to the hallway at angle $-s$ with inner corner $\rho c(s)$, we have $\bigcap_{t \in [-\pi/2, 0]} H_t = \rho S$ and hence

$$\Sigma = \bigcap_{t \in [-\pi/2, \pi/2]} H_t, \qquad c(-t) = \rho c(t). \tag{15}$$

Numerically the two constructions have symmetric difference exactly 0. So the ambidextrous constraint family is a *single* family spanning an angle range of π, rather than two families of range $\pi/2$; and its corner path is discontinuous at $t = 0$, where $c_y(0) = 0$ while $\rho c(0)$ has height 1. Both features lie outside the setting of the one-corner argument, which is developed for rotation angle $\omega \le \pi/2$ and whose maximisers attain $\omega = \pi/2$. Equation (15) is thus a reformulation, not a reduction: it exhibits the ambidextrous problem inside a larger class rather than inside the solved one.

8 The niche area in convex-linear data

The obstruction recorded in Remark 17 is that the one-corner argument's injectivity condition fails for Σ on the two outer phases. We now show that the failure is an artefact of how the condition is stated, and derive an exact formula for the niche area whose every ingredient is convex-linear in the cap.

Throughout, K is the cap, $h = h_K$ its support function, and

$$F(t) := h(\mu_t), \qquad G(t) := h(\nu_t).$$

Lemma 18 (The corner is affine in the support function)**.** *Place the hallway at angle t and push both outer walls inward until they touch K. Then the inner corner sits at*

$$c(t) = \big(F(t) - 1\big)\mu_t + \big(G(t) - 1\big)\nu_t, \tag{16}$$

and the removed wedge is $Q_t = \{\langle p, \mu_t \rangle < F(t) - 1\} \cap \{\langle p, \nu_t \rangle < G(t) - 1\}$.

Proof. The two outer walls are the lines with outer normals μ_t, ν_t; touching K means their offsets are $F(t), G(t)$. The corner lies one unit inside each, so $\langle c, \mu_t\rangle = F - 1$ and $\langle c, \nu_t\rangle = G - 1$, which is (16) since (μ_t, ν_t) is orthonormal. The wedge is the intersection of the two inner half-planes through c. $\square$

Since $h \mapsto h$ is linear under Minkowski sum, (16) makes $c(t)$, and hence each of

$$\alpha_1(t) := -\langle c'(t), \mu_t\rangle = G(t) - 1 - F'(t), \tag{17}$$

$$\alpha_2(t) := \langle c'(t), \nu_t\rangle = F(t) - 1 + G'(t), \tag{18}$$

$$\sigma(t) := c_y(t)/\cos t = \big(F(t) - 1\big)\tan t + G(t) - 1, \tag{19}$$

an affine-linear functional of h, hence convex-linear on a domain of convex bodies closed under Minkowski sum. The wedge Q_t has two faces, the rays $c - s\nu_t$ (face 1) and $c - s\mu_t$ (face 2), and α_1, α_2 are the signed distances from c to the envelope points on them: the one-corner injectivity condition is exactly $\alpha_1, \alpha_2 > 0$.

Lemma 19 (Normal velocities)**.** *The face-*1 *line advances in its own normal direction with speed* $s - \alpha_1(t)$ *at the point at distance* s *from the corner, and the face-*2 *line with speed* $\alpha_2(t) - s$.

Proof. Face 1 lies on $\ell_t = \{\langle p, \mu_t\rangle = F(t) - 1\}$. For p fixed the signed gap to ℓ_t is $F(t) - 1 - \langle p, \mu_t\rangle$, whose t-derivative is $F'(t) - \langle p, \nu_t\rangle$ because $\mu_t' = \nu_t$. At $p = c - s\nu_t$ we have $\langle p, \nu_t\rangle = G - 1 - s$, so the speed is $F' - G + 1 + s = s - \alpha_1$ by (17). Face 2 lies on $\{\langle p, \nu_t\rangle = G(t) - 1\}$ and $\nu_t' = -\mu_t$, so the same computation gives $G' + \langle p, \mu_t\rangle = G' + F - 1 - s = \alpha_2 - s$ at $p = c - s\mu_t$. $\square$

So the niche is created by the *inner* part of face 2 and the *outer* part of face 1, and *no sign hypothesis on* α_1 *is involved*: where $\alpha_1 < 0$ the whole visible face 1 advances, and where $\alpha_2 \le 0$ face 2 contributes nothing. This is the promised repair of Remark 17: each face is used only on the range where its own arm has the right sign, and the reported failure came from requiring both arms simultaneously.

Write W_2 for the region swept by the segments $[c(t), c(t) - \alpha_2(t)^+\mu_t]$, and W_1^{out} for the region swept by the face-1 segments running from $c(t) - \alpha_1(t)^+\nu_t$ outward to the corridor floor $\{y = 0\}$; note the face-1 direction $-\nu_t = (\sin t, -\cos t)$ points downward, and the floor is reached at distance exactly $\sigma(t)$, which is why (19) is the relevant reach.

Proposition 20 (Niche area in convex-linear data)**.** *If the two sweeps are disjoint, injective and cover the niche, then*

$$|N| = \int_0^{\pi/2} \Big[\tfrac{1}{2}\big(\alpha_2^+\big)^2 + \tfrac{1}{2}\big(\sigma - \alpha_1\big)^2 - \tfrac{1}{2}\big(\alpha_1^-\big)^2\Big]\, dt. \tag{20}$$

Proof. For the face-2 sweep $\Phi(s,t) = c(t) - s\mu_t$ one has $\partial_s\Phi = -\mu_t$ and $\partial_t\Phi = c' - s\nu_t$, so $\det[\partial_s\Phi, \partial_t\Phi] = -\langle c', \nu_t\rangle + s = s - \alpha_2$, and integrating $|s - \alpha_2|$ over $s \in [0, \alpha_2^+]$ gives $\frac{1}{2}(\alpha_2^+)^2$. For the face-1 sweep $\Psi(s,t) = c(t) - s\nu_t$ one gets $\det = s - \alpha_1$, and

$$\int_{\alpha_1^+}^{\sigma} (s - \alpha_1)\, ds = \tfrac{1}{2}(\sigma - \alpha_1)^2 - \tfrac{1}{2}(\alpha_1^-)^2,$$

which is $\frac{1}{2}(\sigma - \alpha_1)^2$ when $\alpha_1 \ge 0$ and $\frac{1}{2}\sigma^2 - \alpha_1\sigma$ when $\alpha_1 < 0$, as required. Injectivity turns each $\int |\det|$ into the area of the image, disjointness makes the two areas add, and covering makes the sum $|N|$. $\square$

Remark 21 (What is measured, and what it gives)**.** For Σ the three hypotheses of Proposition 20 are verified numerically and sharply: the two sweeps have intersection area 0 and leave 0 of the niche uncovered, to the printing precision of the polygon oracle. The right-hand side of (20), evaluated with exact derivatives of SOL1/SOL5/SOL6 and Gauss–Legendre quadrature on each phase separately, is

$$V = 0.184193197089\ldots,$$

stable in the twelfth digit from 20 to 320 nodes per phase, against a polygonal measurement 0.184193171 of the same region — and an inscribed quadrilateral union under-measures, so the two agree to $2.6 \cdot 10^{-8}$ from the expected side. The implied cap area $A_R^* + 2V = 2.0133416\ldots$ lies just below the circumscribed polygonal values 2.0133455 and 2.0133420 at 241 and 481 constraints, which must decrease to it. We therefore record (20) as an identity for Σ at the 10^{-8} level.

Two of the three terms of (20) are convex quadratics in h, since $x \mapsto \frac{1}{2}(x^+)^2$ and $x \mapsto \frac{1}{2}x^2$ are convex and $\alpha_2, \sigma - \alpha_1$ are affine in h. This is what the generalised Mamikon theorem buys, and it is the reason (20) is the right shape: for $|\Sigma| = |C_2| - 2|N|$ a convex $|N|$ gives a concave upper bound, for which a first-order condition suffices.

Proposition 22 (The obstruction, in closed form)**.** *The third term of* (20) *is subtracted, so it contributes a* convex *term to* $|C_2| - 2|N|$ *and is the one ingredient with the wrong curvature. It is supported exactly on the degenerate phase* $[0, \beta)$*, where* $\alpha_1 < 0$*, and there* $\alpha_1(t) = 2a_1 \sin t - \frac{1}{2}$*, so*

$$\frac{1}{2}\int_0^{\pi/2} \left(\alpha_1^-\right)^2 dt = \frac{\beta}{8} - a_1(1 - \cos\beta) + a_1^2\left(\beta - \frac{1}{2}\sin 2\beta\right) = 0.011950270059\ldots, \tag{21}$$

which is $6.488\ldots\%$ *of* V*.*

Proof. On $[0, \beta)$ the path is SOL1 with $a_2 = 0$, so in complex form $z(t) = a_1 e^{2it} - (1+\frac{i}{2})e^{it} + (1 - a_1 + \frac{i}{2})$ and $e^{-it}z'(t) = 2ia_1e^{it} - i(1+\frac{i}{2})$, whence $\alpha_1 = -\operatorname{Re}(e^{-it}z') = 2a_1 \sin t - \frac{1}{2}$. This is negative exactly for $\sin t < 1/(4a_1)$, i.e. for $t < \beta$ by Theorem 14. Expanding $\frac{1}{2}\int_0^\beta (\frac{1}{2} - 2a_1 \sin t)^2 dt$ and using $\int_0^\beta \sin^2 = \frac{\beta}{2} - \frac{1}{4}\sin 2\beta$ gives (21). ☐

Remark 23 (Where this leaves the problem)**.** Two obstructions have been removed and one remains. The first was that no trial underestimate of the niche was tight: the best previous one, built from the apex path, left a slack of 0.087 against $|\Sigma| = 1.645$. Formula (20) is tight, so that slack is gone. The second was that the injectivity condition fails for Σ on $[0, \beta)$ and $(\pi/2 - \beta, \pi/2]$; Lemma 19 shows the condition is not needed, only the two one-sided statements α_2^+ and $\sigma \geq \alpha_1^+$, and the second holds throughout with equality only at $t = \pi/2$.

What remains is (21). It is a subtracted square of a convex-linear quantity, i.e. a $\mathcal{J}$-type term of the same kind the one-corner argument must itself control, and it is localised on the phase where Σ degenerates. Establishing or refuting concavity of $|C_2| - 2V$ is a Hessian computation on the space of support-function perturbations, and is not attempted here. Neither is the second half of the argument, which would have to show that the three hypotheses of Proposition 20 hold for competitors and not only for Σ. So *optimality of* Σ *remains open*; what is new is that the upper bound is now tight and expressed entirely in convex-linear data, with a single explicitly computed term of the wrong sign.

9 The upper bound at Σ: tight, critical, concave

Since $\nu_t = \mu_{t+\pi/2}$, everything above is a functional of the single function

$$H(\theta) := h_K(\mu_\theta), \qquad \theta \in [0,\pi], \qquad F(t) = H(t), \quad G(t) = H(t+\pi/2).$$

Proposition 24 (The cap is an exact quadratic form). *Let $C = \bigcap_{t\in[0,\pi/2]} C_t$ and $C_2 = C \cap \rho C$. Then*

$$|C_2| = \int_0^\pi \big(H(\theta)^2 - H'(\theta)^2\big)d\theta - H(0) - H(\pi). \tag{22}$$

Proof. ρ is a reflection in the line $y = \frac{1}{2}$, not in the origin: $\rho p = Rp + (0,1)$ with $R = \operatorname{diag}(1,-1)$, so $h_{\rho A}(u) = h_A(Ru) + u_y$. Since C is unbounded downward, $h_C = +\infty$ on the lower half-circle and $h_{\rho C} = +\infty$ on the upper one, so the support function of C_2 is

$$h_2(\theta) = H(\theta) \ \ (\theta \in [0,\pi]), \qquad h_2(\theta) = H(-\theta) + \sin\theta \ \ (\theta \in [-\pi, 0]).$$

For any support function in H^1 of the circle, $|C_2| = \frac{1}{2}\int h_2\, d\sigma_2$ with $\sigma_2 = (h_2 + h_2'')d\theta$ as a measure, and the distributional identity $\langle h_2'', h_2\rangle = -\int h_2'^2$ holds with no boundary or atom terms because the circle has no boundary. Splitting at $\theta = 0, \pi$ and substituting the two branches,

$$|C_2| = \int_0^\pi \Big[H^2 - H'^2 - H\sin s + H'\cos s - \tfrac{1}{2}\cos 2s\Big]ds,$$

and $\int_0^\pi \cos 2s\, ds = 0$ while $\int_0^\pi H'\cos s\, ds = -H(0) - H(\pi) + \int_0^\pi H \sin s\, ds$, which gives (22). $\square$

Note $-H(0) - H(\pi)$ is *linear*, so it does not enter the Hessian; and (22) is invariant under $H \mapsto H + v\cos\theta$, as it must be, that being a horizontal translation. Numerically (22) gives 2.013341613 against $A_R^* + 2V = 2.013341613$, agreeing to $9.6\cdot 10^{-13}$ — an independent analytic confirmation of the whole chain $|\Sigma| = |C_2| - 2|N|$ with $|N| = V$, since (22) and (20) share no computation. Equivalently, writing $Q := |C_2| - 2V$,

$$Q(\Sigma) = 1.644955218425\ldots = A_R^* \quad \text{to } 5\cdot 10^{-13}. \tag{23}$$

Proposition 25 (The second variation). *Fix the gauge $H(0) = 1$ (horizontal translation) and $H(\pi/2) = 1$ (unit corridor; this is also exactly what makes σ integrable against $\tan t$ at $t = \pi/2$), so that admissible η satisfy $\eta(0) = \eta(\pi/2) = 0$. Then, with $E_1 = \{\alpha_1 < 0\}$ and $E_2 = \{\alpha_2 > 0\}$,*

$$\begin{aligned}\tfrac{1}{2}\,\delta^2 Q[\eta] = \int_0^\pi (\eta^2 - \eta'^2)\,d\theta &- \int_{E_2}\big(\eta(t) + \eta'(t+\tfrac{\pi}{2})\big)^2 dt \\ &- \int_0^{\pi/2}\big(\eta(t)\tan t + \eta'(t)\big)^2 dt + \int_{E_1}\big(\eta(t+\tfrac{\pi}{2}) - \eta'(t)\big)^2 dt.\end{aligned} \tag{24}$$

Its principal part is negative definite: the coefficient of $\eta'(\theta)^2$ is -1 on $[0,\beta)$ and on $[\pi-\beta,\pi]$, and -2 in between. Hence (24) is bounded above and has at most finitely many non-negative eigenvalues.

Proof. $\delta\alpha_1 = \eta(t+\frac{\pi}{2}) - \eta'(t)$, $\delta\alpha_2 = \eta(t) + \eta'(t+\frac{\pi}{2})$ and $\delta\sigma = \eta(t)\tan t + \eta(t+\frac{\pi}{2})$ by (17)–(19); in the middle term the $\eta(t+\pi/2)$ cancels, leaving $\delta(\sigma - \alpha_1) = \eta(t)\tan t + \eta'(t)$. The integrand of (20) is C^1 across $\alpha_1 = 0$ and $\alpha_2 = 0$, so the moving boundaries of E_1, E_2 contribute nothing at second order, and δ^2 of each square is twice the square of the first variation. The cap contributes $2\int(\eta^2 - \eta'^2)$ by Proposition 24. For the principal part, collect: -1 from the cap, -1 from the σ term on $[0,\pi/2]$, $+1$ from the E_1 term on $[0,\beta)$, and -1 from the E_2 term at $\theta = t + \pi/2 \in [\pi/2, \pi-\beta)$. $\square$

Remark 26 (Measurements, and exactly what they do and do not give)**.** In a hat basis with $\eta(0) = \eta(\pi/2) = 0$:

Criticality. $\delta Q[\eta] = 0$ in every direction, $\|\delta Q\|_\infty/h = 1.8 \cdot 10^{-8}$ by central differences on Q itself. So Romik's ODEs are the Euler–Lagrange equations of Q. (An analytic assembly of the gradient first reported a nonzero value on $[0, \beta)$ and $[\pi/2, \pi/2 + \beta)$; that was a sign error on $\delta\big(-\frac{1}{2}(\alpha_1^-)^2\big) = +\alpha_1^- \delta\alpha_1$, caught by the finite-difference check.)

Concavity, on Σ's cell only. With Σ's own sign pattern $E_1 = [0, \beta)$, $E_2 = [0, \pi/2 - \beta)$, the form (24) is negative definite: $\sup \frac{1}{2}\delta^2 Q[\eta]/\|\eta\|^2_{L^2(0,\pi)} = -0.7358, -0.7339, -0.7331$ at $m = 32, 64, 128$, against the exact mass matrix. Under Baek's injectivity condition, which is exactly $E_1 = \emptyset$, $E_2 = [0, \pi/2]$, it is -0.8751. But (24) is *not* negative for every sign pattern: with $E_1 = E_2 = [0.4, 1.2]$ it is $+0.4123$, and with $E_1 = E_2$ a union of three intervals $+0.4225$. An earlier scan over intervals *anchored at* 0 suggested that $\tau_1 \le \tau_2$ suffices; the unanchored cases refute that and the suggestion is **retracted**.

Since α_1, α_2 are affine in H, each pointwise sign condition is a half-space and Σ's cell is convex. Concavity plus criticality on a convex set gives that Σ maximises Q *on that cell.* Nothing global follows.

What is missing — and it is worse than missing. All of the above concerns Q alone. The inequality $Q \ge |\text{sofa}|$, which is the only reason Q is relevant, **fails**; see Proposition 28. All numbers in this remark are floating point.

Remark 27 (The boundary of the cap, and where the sweep terminates)**.** The density $H + H''$ of the surface measure determines ∂C_2 completely: it vanishes on $(-\beta, \beta)$, where ∂C_2 is the single vertex $P = (1, \frac{1}{2})$; equals $\frac{1}{2}$ on $(\pi/2 - \beta, \pi/2 + \beta)$, where ∂C_2 is a circular arc of radius exactly $\frac{1}{2}$ centred at $(1 - 2a_1, \frac{1}{2})$; has an atom of mass 1.167050 at $\theta = \pi/2$, the corridor ceiling $y = 1$, a facet; vanishes again on $(\pi - \beta, \pi]$, the second vertex; and is a smooth arc in between. The ρ-image of the ceiling facet is the corridor floor, the segment $[-0.750575, 0.416475] \times \{0\}$, whose left endpoint lies directly below the centre of the radius-$\frac{1}{2}$ arc.

This settles the geometry of the outer arm of Proposition 20. Because C_2 is *convex*, the face-1 segment lies in C_2 as soon as its far endpoint does, so the claim $S_1 = \sigma$ reduces to a one-parameter containment: the endpoint $x(t) = c_x(t) + \sigma(t)\sin t$ must lie in the floor facet. It runs monotonically from $x(0) = 0$ to $x(\pi/2) = 0.416475$, the right end of the facet, and

$$x(\pi/2) = c_x(\pi/2) + \alpha_1(\pi/2) = \big(c_x(0) - \alpha_2(0)\big) + \big(\alpha_2(0) + \alpha_1(\pi/2) - (G(\pi/2) - 1)\big)$$

is an algebraic identity, the right-hand side being the left end of the facet plus its length $H'(\pi/2^+) - H'(\pi/2^-)$. So the tangency at $t = \pi/2$ is exact, not coincidental, and what remains of $S_1 = \sigma$ is the monotonicity of the explicit function x.

Proposition 28 (V over-estimates the niche, so Q is not an upper bound)**.** *For every admissible H, $V(H) \ge |N(H)|$. Consequently $Q = |C_2| - 2V \le |C_2| - 2|N|$, and for the maximal sofa $T_{\max} = C_2 \setminus (U \cup \rho U)$ with that cap, $|T_{\max}| = |C_2| - 2|N| \ge Q$. So Q is an upper bound for $|T_{\max}|$ only where $V = |N|$.*

Proof. V is precisely the flux of the advancing part of ∂Q_t, by Lemma 19 and the Jacobian computation in Proposition 20. By Reynolds' transport theorem the area gained by the sweeping region $N_T = \bigcup_{t \le T} Q_t \cap C_2$ satisfies $\frac{d}{dT}|N_T| \le \int_{\text{advancing part of } \partial Q_T} v_n \, ds$, because advance into already-swept territory contributes to the flux but gains no area. Integrating over $T \in [0, \pi/2]$ gives $|N| \le V$. □

Remark 29 (The size of the gap, measured)**.** Equality in Proposition 28 holds exactly when nothing is re-swept, which is what the three hypotheses of Proposition 20 assert, and which is why $V(\Sigma) = |N(\Sigma)|$. Perturbing H by a smooth bump η supported where $H + H'' \geq \frac{1}{2}$ (so convexity survives) and measuring $|N|$ with a polygon oracle against V from the same H, the excess $V - |N|$, corrected for the oracle's known $O(1/n)$ bias, is

ε	-0.20	-0.10	-0.05	0.05	0.10	0.20
$V - \|N\|$	$5.5 \cdot 10^{-3}$	$1.4 \cdot 10^{-4}$	$4.2 \cdot 10^{-6}$	$9.3 \cdot 10^{-5}$	$8.4 \cdot 10^{-4}$	$5.0 \cdot 10^{-3}$

— strictly positive on *both* sides and growing faster than ε^2. So Σ is an isolated zero of $V - |N|$: the functional Q touches the true bound at Σ and falls strictly below it in every direction. Therefore Q cannot certify Σ, and the fact that Q is tight, critical and concave there (Remark 26) does not yield an optimality statement.

This locates the difficulty exactly. The repair is the one the one-corner argument performs: replace the exact flux V by the flux over a sub-family on which the sweep is *provably* injective, so that the integral is the area of a genuine subset of N and the inequality runs the right way, while the restriction is vacuous at Σ. Producing such a sub-family is the content of that argument's Chapters 3–6 and is not done here.

One reformulation that may help. A point p lies on the face-2 line at parameter t iff $\phi(t) := \langle p, \nu_t \rangle - (G(t) - 1)$ vanishes, and with $s(p,t) := F(t) - 1 - \langle p, \mu_t \rangle$ one computes $\phi' = s - \alpha_2$ and $s' = -\phi - \alpha_1$, i.e. in complex form

$$z := s + i\phi, \qquad z' = iz - (\alpha_1 + i\alpha_2).$$

So the whole face-2 sweep is governed by one linear ODE whose homogeneous part is an exact rotation, $z(t) = \zeta(t) + e^{it} w_0$ with w_0 an affine coordinate for p; the sweep is injective iff no solution meets $\{\phi = 0,\ 0 \leq s \leq \alpha_2\}$ twice, and at every such meeting $\phi' = s - \alpha_2 \leq 0$, so the crossing is always downward. A second meeting therefore forces an upward crossing in between, at $s \geq \alpha_2$. Since the total rotation over $[0, \pi/2]$ is only $\pi/2$, the homogeneous part alone cannot supply that extra crossing; whether the forcing ζ can is the open point.

Proposition 30 (Monotonicity of the outer arm)**.** *The face-1 segment's far endpoint has x-coordinate*

$$x(t) = c_x(t) + \sigma(t) \sin t = \frac{F(t) - 1}{\cos t}, \tag{25}$$

the x-intercept of the face-1 line, and $\cos^2 t\, x'(t) = F'(t) \cos t + (F(t) - 1) \sin t$*. For* Σ*:*

- *on* $[0, \beta)$*, where* $A(t) \equiv P$ *so* $F(t) = \cos t + \frac{1}{2} \sin t$*,*

  $$x = 1 + \tfrac{1}{2} \tan t - \sec t, \qquad \cos^2 t\, x' = \tfrac{1}{2} - \sin t > 0,$$

  the last because $\sin \beta = 1/(4a_1) = 0.2856\ldots < \frac{1}{2}$ *by Theorem 14, i.e.* $\beta < \pi/6$*;*

- *on* $(\pi/2 - \beta, \pi/2]$*, where* $F(t) - 1 = A \cos t + \frac{1}{2} \sin t - \frac{1}{2}$ *with* $A = 1 - \frac{2}{3} a_1$*,*

  $$x = A - \tfrac{1}{2} \tan\Big(\tfrac{\pi}{4} - \tfrac{t}{2}\Big), \qquad \cos^2 t\, x' = \tfrac{1}{2}(1 - \sin t) > 0 \quad (t < \pi/2),$$

  and in particular $x(\pi/2) = A = 1 - \frac{2}{3} a_1 = 0.4164750917\ldots$*, which is exactly the right endpoint of the corridor floor facet of Remark 27;*

- *on $[\beta, \pi/2-\beta]$, substituting $u = t/2 - \pi/8$ and using $-f_2/f_1 = \sqrt{2}-1 = \tan(\pi/8)$,*

$$\cos^2 t\, x' = W(u) := \tfrac{3K}{4}\sin u + \tfrac{K}{4}\cos 3u + \tfrac{1}{2} - \sin\Big(2u + \tfrac{\pi}{4}\Big), \qquad K = f_1\sqrt{4-2\sqrt{2}},$$

on $|u| \le \pi/8 - \beta/2$.

Hence x is strictly increasing, given $W > 0$.

Proof. The face-1 line is $\{\langle p, \mu_t\rangle = F(t) - 1\}$, so its x-intercept is $(F-1)/\cos t$; that it agrees with $c_x + \sigma \sin t$ is $c_x + c_y \tan t = (F-1)/\cos t$, immediate from $c = (F-1)\mu_t + (G-1)\nu_t$. The three closed forms follow by substituting $F(t) - 1 = \mathrm{Re}(e^{-it}z(t))$ with the complex forms of SOL1, SOL6, SOL5. For the middle phase the κ_6 terms cancel and the triple-angle identities $\sin(3t/2) = s(3c^2 - s^2)$, $\cos(3t/2) = c(c^2 - 3s^2)$ with $s = \sin\frac{t}{2}$, $c = \cos\frac{t}{2}$ reduce the result to the stated three-term form. □

Proposition 31 ($W > 0$, by a finite certificate)**.** *$W > 0$ on $|u| \le u_0 := \pi/8 - \beta/2 = 0.2478721712\ldots$, so x is strictly increasing on all of $[0, \pi/2]$ and $S_1 = \sigma$.*

Proof. $K = 1.3020516916172893452\ldots$, and $3u_0 = 0.7436\ldots < \pi$, so $\cos 3u$ is unimodal on $[-u_0, u_0]$ with maximum at 0 and its minimum over any subinterval is attained at an endpoint. Also $\sin u$ is increasing, and $2u+\pi/4$ ranges over $[0.2896\ldots, 1.2812\ldots] \subset [0, \pi/2]$, so $\sin(2u+\pi/4)$ is increasing. Therefore on $[p,q] \subseteq [-u_0, u_0]$,

$$W \;\ge\; L(p,q) := \tfrac{3K}{4}\sin p + \tfrac{K}{4}\min\big(\cos 3p, \cos 3q\big) + \tfrac{1}{2} - \sin\Big(2q + \tfrac{\pi}{4}\Big).$$

Dividing $[-u_0, u_0]$ into 32 equal pieces gives $L > 0$ on every piece, with $\min L = 6.13 \cdot 10^{-3}$; the computation is at 40 working digits, so the margin exceeds the arithmetic error by 37 orders of magnitude. (16 pieces do not suffice: $\min L = -8.6 \cdot 10^{-3}$.) □

Remark 32. *W is in fact strictly decreasing, with $W(u_0) = \frac{1}{2}(1-\cos\beta) = 0.020828595599846555\ldots$ at the right junction, matching the last phase's $\frac{1}{2}(1-\sin t)$ there to $1.7 \cdot 10^{-41}$, as C^1 gluing requires. The three closed forms of Proposition 30 were also checked against numerical differentiation of the path to 10^{-10}.*

Remark 33 (The identity at 61 digits)**.** With all constants from their closed forms ($u^3+3u = 2$, $\beta = \arctan(u/2)$, $a_1 = \sqrt{4+u^2}/(4u)$, f_1 from Proposition 6) and 60 working digits,

$$V = 0.184193197088768988251655564606\ldots, \qquad |C_2| = 2.013341612602978828172120476813\ldots,$$

$$Q(\Sigma) - A_R^* = -1.56 \cdot 10^{-61}.$$

The two sides share no computation, so this rules out the possibility that the double-precision agreement of (23) was a rounding coincidence. It remains floating point, not an interval enclosure.

10 A convex subdomain on which the bound is valid

Proposition 28 says $V \geq |N|$ with equality exactly when nothing is re-swept, i.e. when the combined sweep is injective. We now show that injectivity is governed by inequalities that are *linear* in H, so the set where the bound is valid is convex.

Each sweep lies on a line: face 2 at parameter t on $\ell_2(t) = \{\langle p, \nu_t\rangle = G(t) - 1\}$, face 1 on $\ell_1(t) = \{\langle p, \mu_t\rangle = F(t) - 1\}$. Two distinct lines meet in one point, so a double cover forces that point into both segments. Solving $\ell_2(t) \cap \ell_2(t')$ for the coordinate s on $\ell_2(t)$, and using $\langle \mu_t, \nu_{t'}\rangle = \sin(t - t')$,

$$s_{22}(t, t') = \frac{\langle c(t), \nu_{t'}\rangle - (G(t') - 1)}{\sin(t - t')}, \qquad s_{11}(t, t') = \frac{\langle c(t), \mu_{t'}\rangle - (F(t') - 1)}{\sin(t' - t)}. \tag{26}$$

The numerator of s_{22} vanishes at $t' = t$ with derivative $-\alpha_2(t)$, so $s_{22}(t, t') \to \alpha_2(t)$: nearby face-2 lines meet at the *far* end of the face-2 segment $[0, \alpha_2]$. Likewise $s_{11}(t, t') \to \alpha_1(t)$, the *near* end of the face-1 segment $[\alpha_1^+, \sigma]$.

Definition 34. Let $\mathcal{K}'$ be the set of admissible H satisfying, for all $0 \leq t' < t \leq \pi/2$,

$$\text{(C22)} \quad \langle c(t), \nu_{t'}\rangle - (G(t') - 1) \;\geq\; \alpha_2(t) \sin(t - t'), \tag{27}$$

$$\text{(C11)} \quad \langle c(t), \mu_{t'}\rangle - (F(t') - 1) \;\leq\; \alpha_1(t)^+ \sin(t' - t) \;\; (\text{read with } \sin(t' - t) < 0), \tag{28}$$

together with the analogous condition for $\ell_2(t) \cap \ell_1(t')$.

Proposition 35. *$\mathcal{K}'$ is convex, and on $\mathcal{K}'$ the combined sweep is injective, hence $V = |N|$ and $Q = |C_2| - 2|N| \geq |T|$ for every ambidextrous sofa T with cap data H.*

Proof. By Lemma 18, $c(t)$ is affine in H, hence so are $\langle c(t), \nu_{t'}\rangle$, $G(t') - 1$, $F(t') - 1$, $\alpha_1(t)$ and $\alpha_2(t)$; $\sin(t - t')$ does not depend on H. So (27) and (28) are linear inequalities in H and cut out a convex set. Given (27), for any pair $t' < t$ the intersection point of $\ell_2(t)$ and $\ell_2(t')$ has $s_{22}(t, t') \geq \alpha_2(t)$, so it is not interior to the segment at t, and the pair contributes no double cover; (28) does the same for face 1 with the inequality reversed because the relevant endpoint is the near one. With the cross condition, no point is swept twice.

For $V = |N|$: every point of N has a first capture time $\tau(p)$, at which $\partial Q_{\tau(p)}$ is advancing at p (otherwise p would have been captured earlier), so by Lemma 19 it lies on face 2 with $s \leq \alpha_2$ or on face 1 with $s \geq \alpha_1$. Hence N is covered by the two sweeps, and the sweeps lie in N because an advancing face point belongs to $Q_{t+\varepsilon}$. Injectivity turns each $\int |\det|$ into the area of its image, and the two images are disjoint, so the sum is $|N|$. $\square$

Remark 36 ($\Sigma \in \mathcal{K}'$, and $\mathcal{K}'$ has nonempty interior)**.** Measured on a grid of 401 parameters, Σ satisfies all three conditions: of the 158 802 ordered pairs with $|t - t'|$ above the diagonal cutoff, *zero* give a point interior to both face-2 segments and *zero* interior to both face-1 segments, and of 58 081 cross pairs, *zero* land in both. The margins in (27), (28) vanish linearly as $t' \to t$, which is forced: that is the envelope limit. Away from the diagonal they are strictly positive.

$\mathcal{K}'$ is not a single point. For $H = H_\Sigma + \varepsilon \sin(k\theta)$, which respects the gauge for even k, all conditions still hold with zero violations at $k = 2$ for $|\varepsilon| \leq 3 \cdot 10^{-2}$ and at $k = 4$ for $|\varepsilon| \leq 10^{-2}$, in both signs, and the two margins trade off smoothly. So Σ lies in $\mathcal{K}'$ with low-frequency room around it.

The mechanism of failure is identified: for the compactly supported bump perturbations of Remark 29, where $V - |N|$ was found positive, the face-2 condition continues to hold and it is the *face-1* condition that fails, by $-4.7 \cdot 10^{-2}$ at $\varepsilon = 0.05$ and $-2.3 \cdot 10^{-1}$ at $\varepsilon = 0.20$. Those perturbations have large higher derivatives, which is what creates face-1 envelope self-intersections.

Theorem 37 (Conditional)**.** *Assume:*

(i) $Q(\Sigma) = A^*_R$*,* $\delta Q(\Sigma) = 0$ *in every admissible direction, and* $\delta^2 Q \preceq 0$ *on the cell* $\mathcal{C}$ *of support functions sharing* Σ*'s sign pattern;*

(ii) the three conditions of Definition 34 hold on $\mathcal{K}' \cap \mathcal{C}$*;*

(iii) competitors satisfy the niche separation $M < \frac{1}{2}$ *of Corollary 2.*

Then $|T| \leq A^*_R$ *for every ambidextrous sofa* T *whose cap data lies in* $\mathcal{K}' \cap \mathcal{C}$*.*

Proof. $\mathcal{C}$ is convex because α_1, α_2 are affine in H, so each pointwise sign condition is a half-space; $\mathcal{K}'$ is convex by Proposition 35; hence so is the intersection. By (i), Q is a concave function on that convex set with a critical point at Σ, so $Q(H) \leq Q(\Sigma) = A^*_R$ throughout. By (ii) and Proposition 35, $Q(H) \geq |T|$ there, using (iii) to get the factor 2 in $|T| \leq |C_2| - 2|N|$. □

Remark 38 (Honest status of Theorem 37)**.** Hypothesis (i) is *numerical*: $Q(\Sigma) = A^*_R$ to 61 digits (Remark 33), $\|\delta Q\|_\infty / h = 1.8 \cdot 10^{-8}$, and $\sup \frac{1}{2}\delta^2 Q / \|\eta\|^2_{L^2} \to -0.733$ for a discretised form. Hypothesis (ii) is measured on a grid, not proved. Hypothesis (iii) is the conditional part of Theorem 11. So the effective label of Theorem 37 is HEURISTIC, and it does not assert optimality of Σ: the class $\mathcal{K}' \cap \mathcal{C}$ is not shown to contain every competitor, and by Remark 29 bodies outside $\mathcal{K}'$ genuinely violate $Q \geq |T|$. What the theorem does is exhibit an explicit convex class, with nonempty interior around Σ, on which the one-corner architecture transfers. Proving (i) and (ii) is the remaining work, and is finite: (i) reduces to a Hessian sign and a first-order identity in closed form, and (ii) to the linear inequalities (27), (28).

11 One curvature condition implies injectivity

The conditions of Definition 34 were verified on a grid. We now prove them, for all pairs at once, from a single linear condition on the cap.

Definition 39. Say H satisfies (RC) if the absolutely continuous part of the surface measure obeys

$$H(\theta) + H''(\theta) \;\leq\; 1 \qquad \text{for a.e. } \theta \in [0, \pi],$$

the only atoms being the corridor ceiling and floor facets at $\theta = \pm\pi/2$. Equivalently: the cap is nowhere flatter than a circle whose radius is the corridor width.

Since $H + H'' \leq 1$ is linear in H, (RC) cuts out a convex set.

Remark 40 ((RC) is classical)**.** The bound $h + h'' \leq R$ is not new. By the dual of Blaschke's rolling theorem, a convex body K slides freely inside a ball of radius R exactly when $h_K + h''_K \leq R$, and that in turn says K is a Minkowski summand of RB; see Schneider [4, §3.2]. Our (RC) is that condition at $R = 1$, the corridor width, *relaxed* to permit atoms at $\theta = \pm\pi/2$. Since

Minkowski addition adds surface area measures, and a body whose surface measure is two equal atoms at $\pm\pi/2$ is a horizontal segment,

$$\text{(RC)} \quad \Longleftrightarrow \quad C_2 = \big[\text{a horizontal segment}\big] \oplus D, \qquad D \text{ a Minkowski summand of the unit disc.}$$

For Σ the segment has length $1.1670498\ldots$, the corridor facet, and D has curvature radius at most 0.8389. What we claim is not the condition but its use: that it forces the niche sweep to be injective (Theorems 41, 45).

Theorem 41. *If H satisfies (RC), then (27) and (28) hold for every pair $0 \le t' < t \le \pi/2$. Hence each of the two sweeps is injective.*

Proof. Fix t and set $\tau = t - t' \in (0, t]$. For (27) put $n(t') = \langle c(t), \nu_{t'}\rangle - (G(t') - 1)$ and

$$\Phi(\tau) = n(t - \tau) - \alpha_2(t)\sin\tau,$$

so that (27) reads $\Phi \ge 0$. Now $n(t) = 0$ because $\langle c(t), \nu_t\rangle = G(t) - 1$, and $n'(t') = -\langle c(t), \mu_{t'}\rangle - G'(t')$ equals $-(F(t) - 1) - G'(t) = -\alpha_2(t)$ at $t' = t$ by (18); hence $\Phi(0) = \Phi'(0) = 0$. Differentiating again and using $\nu'' = -\nu$,

$$n''(t') = -\langle c(t), \nu_{t'}\rangle - G''(t') = -\big(n(t') + G(t') - 1\big) - G''(t'),$$

so with $n(t - \tau) = \Phi(\tau) + \alpha_2(t)\sin\tau$ the α_2 terms cancel and

$$\Phi''(\tau) + \Phi(\tau) = 1 - (G + G'')(t - \tau) =: R(\tau). \tag{29}$$

With $\Phi(0) = \Phi'(0) = 0$, (29) gives $\Phi(\tau) = \int_0^\tau \sin(\tau - u)R(u)\,du$. Since $\tau \le t \le \pi/2 < \pi$ we have $\sin(\tau - u) \ge 0$ for $u \in [0, \tau]$, and $R \ge 0$ by (RC) because $G(s) = H(s + \pi/2)$. Therefore $\Phi \ge 0$.

The ceiling atom of H sits at $\theta = \pi/2$, i.e. at $t - u + \pi/2 = \pi/2$, i.e. $u = t$. For $\tau < t$ that is outside $[0, \tau]$, and for $\tau = t$ it is the endpoint, where the kernel $\sin(\tau - u)$ vanishes. So the atom contributes nothing.

For (28) put $m(t') = \langle c(t), \mu_{t'}\rangle - (F(t') - 1)$ and $\Psi(\tau) = m(t - \tau) + \alpha_1(t)\sin\tau$. Then $m(t) = 0$, $m'(t) = \alpha_1(t)$ by (17), so $\Psi(0) = \Psi'(0) = 0$, and $\mu'' = -\mu$ gives in the same way

$$\Psi''(\tau) + \Psi(\tau) = 1 - (F + F'')(t - \tau),$$

whence $\Psi(\tau) = \int_0^\tau \sin(\tau - u)\big[1 - (H + H'')(t - u)\big]\,du \ge 0$. Multiplying by $-\sin\tau < 0$ turns $\Psi \ge 0$ into $s_{11} \le \alpha_1$, which is (28). Here the atom needs $t - u = \pi/2$, i.e. $u = t - \pi/2 \le 0$, reachable only at $t = \pi/2$, where $\sigma(\pi/2) = \alpha_1(\pi/2)$ makes the face-1 segment empty and the condition vacuous. $\square$

Remark 42 (Verification)**.** For Σ the absolutely continuous part of $H + H''$ has maximum $0.8385682\ldots$, at $\theta = \pi - \beta$, so (RC) holds with margin $0.1614\ldots$; the atom at $\theta = \pi/2$ has mass $1.1670498\ldots$, the ceiling facet length. The reduction (29) was checked against direct evaluation of $n(t - \tau) - \alpha_2(t)\sin\tau$ at several (t, τ), agreeing to $2.4 \cdot 10^{-5}$, the finite-difference floor for H''.

Theorem 41 also explains the failure mode of Remark 36. Across twelve perturbations, (RC) holds exactly when (28) holds: the compactly supported bumps give $\max(H + H'')_{\mathrm{ac}} =$ 1.08, 1.45, 2.18, 2.67, 4.58, 8.41 and all violate (28); $H_\Sigma + \varepsilon\sin(2\theta)$ at $|\varepsilon| = 0.03$ gives 0.888 and holds; $H_\Sigma + \varepsilon\sin(4\theta)$ gives 0.976 at $|\varepsilon| = 0.01$ and holds, 1.267 at $|\varepsilon| = 0.03$ and fails. The agreement is exact on both sides of the transition.

Remark 43 (What Theorem 41 does and does not give)**.** It replaces hypothesis (ii) of Theorem 37, for the two self-intersection conditions, by (RC): a single explicit linear inequality with a geometric reading. The cross condition, that $\ell_2(t) \cap \ell_1(t')$ avoid both segments, is not covered; it is measured to hold at Σ (0 of 58 081 pairs) and remains a hypothesis. So the hypothesis class of Theorem 37 may be replaced by $\{(\mathrm{RC})\} \cap \{\mathrm{cross}\} \cap \mathcal{C}$, an intersection of two convex sets with one measured condition.

Remark 44 (The first variation, pointwise)**.** Hypothesis (i) of Theorem 37 contains $\delta Q(\Sigma) = 0$. Integrating every η' in δQ by parts and collecting the coefficient of $\eta(\theta)$ gives the Euler–Lagrange equations of Q:

$$H + H'' = \alpha_2^+ + (\sigma - \alpha_1)\tan\theta - (\sigma - \alpha_1)' + (\alpha_1^-)' \qquad (0 < \theta < \pi/2),$$

$$H + H'' = -(\alpha_2^+)'(s) + \alpha_1^-(s), \qquad s = \theta - \pi/2 \qquad (\pi/2 < \theta < \pi).$$

Evaluated on Σ at eighteen values of θ spanning all six regions, the residual is at most $9.4 \cdot 10^{-6}$, which is the finite-difference floor. So $\delta Q(\Sigma) = 0$ has an explicit pointwise form; proving it amounts to substituting the closed forms of SOL1, SOL5, SOL6 phase by phase, and is not carried out here.

12 The cross condition, the first variation, and a Gårding bound

Theorem 45. *(RC) implies the cross condition: for all t, t' the point $\ell_2(t) \cap \ell_1(t')$ is not interior to both sweep segments. Consequently, under (RC) the combined sweep is injective and $V = |N|$, so $Q = |C_2| - 2|N| \geq |T|$.*

Proof. Work in the frame at t and write $p = c(t) + a\mu_t + b\nu_t$. The equation $\langle p, \nu_t\rangle = G(t) - 1$ forces $b = 0$, so p moves along $\ell_2(t)$ with $s_2 = -a$; the equation $\langle p, \mu_{t'}\rangle = F(t') - 1$ then gives, for $t' = t - \tau$,

$$s_2 = \frac{m(t-\tau)}{\cos\tau}, \qquad s_1 = -\Phi(\tau) - \big(\alpha_2(t) - s_2\big)\sin\tau,$$

using $\langle \mu_t, \nu_{t-\tau}\rangle = \sin\tau$ and $n(t-\tau) = \Phi(\tau) + \alpha_2(t)\sin\tau$. If $\tau > 0$ and p is interior to the face-2 segment, then $0 < s_2 < \alpha_2(t)$, and $\Phi \geq 0$ by Theorem 41 with $\sin\tau \geq 0$ because $\tau \leq \pi/2$; hence $s_1 < 0 \leq \alpha_1(t')^+$ and p lies below the face-1 segment.

For $t' > t$ work in the frame at t': the same computation with the roles of the two families exchanged gives, for $t = t' - \tau'$ with $\tau' > 0$,

$$s_2 = -\Psi(\tau') - \big(s_1 - \alpha_1(t')\big)\sin\tau',$$

so if p is interior to the face-1 segment then $s_1 > \alpha_1(t')^+ \geq \alpha_1(t')$ and $s_2 < 0$, outside $[0, \alpha_2(t)]$. At $t' = t$ the two lines are perpendicular and meet at $c(t)$, where $s_1 = s_2 = 0$, interior to neither. Both displayed identities were checked against a direct linear solve, agreeing to 10^{-16}. □

So hypothesis (ii) of Theorem 37 may be replaced throughout by (RC).

Theorem 46. $\delta Q(\Sigma) = 0$. *Two things are needed and both hold: the Euler–Lagrange equations of Remark 44 hold identically on each of the six regions, as trigonometric identities in a_1, f_1, f_2 with no relation among the constants used; and the boundary term of δQ vanishes, because $H'(\pi) = -\frac{1}{2}$.*

Proof. On each phase, $F(t)-1=\operatorname{Re}(e^{-it}z(t))$ and $G(t)-1=\operatorname{Im}(e^{-it}z(t))$ evaluate to

$$\begin{array}{lll}
\text{SOL1} & F-1=\cos t+\tfrac{1}{2}\sin t-1, & G-1=(2a_1-1)\sin t+\tfrac{1}{2}\cos t-\tfrac{1}{2}, \\
\text{SOL6} & F-1=f_1\cos\tfrac{t}{2}+f_2\sin\tfrac{t}{2}-1+\kappa\cos t+\tfrac{1}{2}\sin t, & G-1=-f_2\cos\tfrac{t}{2}+f_1\sin\tfrac{t}{2}-1+\tfrac{1}{2}\cos t-\kappa\sin t, \\
\text{SOL5} & F-1=(1-\tfrac{2}{3}a_1)\cos t+\tfrac{1}{2}\sin t-\tfrac{1}{2}, & G-1=(\tfrac{8}{3}a_1-1)\sin t+\tfrac{1}{2}\cos t-1,
\end{array}$$

with $\kappa=1-\frac{4}{3}a_1$. Substituting these into $\alpha_1=G-1-F'$, $\alpha_2=F-1+G'$, $\sigma=(F-1)\tan t+G-1$ and into the two Euler–Lagrange equations, with the sign pattern of each region fixed ($\alpha_2>0$ except on the last phase, $\alpha_1<0$ only on the first), every residual reduces identically to 0.

For the boundary term, integrating the η' of $\delta|C_2|=2\int_0^\pi(H\eta-H'\eta')d\theta-\eta(0)-\eta(\pi)$ by parts and using $\eta(0)=0$ leaves $-(2H'(\pi)+1)\eta(\pi)$ beyond the pointwise equations. The variation δV contributes nothing there: its only boundary term at $\theta=\pi$ carries the factor $\alpha_2^+(\pi/2)$, and $\alpha_2(\pi/2)<0$, so the clamp kills it; every other boundary sits at $\theta=0$ or $\pi/2$, where η vanishes. Finally the boundary point of C_2 with outer normal $\mu_\pi=(-1,0)$ is $H(\pi)\mu_\pi+H'(\pi)\nu_\pi=(-H(\pi),-H'(\pi))$, so its height is $-H'(\pi)$; since C_2 is ρ-symmetric about $y=\frac{1}{2}$ its leftmost point is ρ-fixed when unique, forcing $H'(\pi)=-\frac{1}{2}$ and the boundary term to vanish. The same mechanism at $\theta=0$ gives $H'(0)=\frac{1}{2}$: both extreme points of the cap sit at height $\frac{1}{2}$. $\square$

Remark 47. The computation is a computer algebra verification, six residuals, each returning 0 with a_1, f_1, f_2 free. It was controlled against five deliberately perturbed variants of the middle-phase equation (sign flip on the tan term, each of the two terms dropped, α_1 and α_2 exchanged, $H+H''$ replaced by $H-H''$), all of which return nonzero; and the three pairs of phase formulas were checked against the reference path to $2.2\cdot10^{-16}$. The content of Theorem 46 is that Romik's ODE families are exactly the critical-point equations of Q.

Remark 48 (Towards $\delta^2Q\preceq 0$)**.** The remaining part of hypothesis (i) is the sign of the second variation. The step that makes it accessible is an exact algebraic cancellation. On $[0,\beta)$ write $p=\eta'(t)$, $q=\eta(t)$, $r=\eta(t+\pi/2)$, $T=\tan t$; the three terms of (24) that carry $\eta'(t)$ combine as

$$-p^2-(qT+p)^2+(r-p)^2=-(p+qT+r)^2+2r^2+2qTr, \tag{30}$$

so the added term is absorbed exactly and the remainder carries no derivative of η. What is left is a lower-order estimate. With $\eta(0)=\eta(\pi/2)=0$ the relevant Poincaré constants are 4 on $[0,\pi/2]$, 1 on $[\pi/2,\pi]$ (Dirichlet–Neumann, $\eta(\pi)$ free) and $(\pi/2\beta)^2=29.41$ on $[0,\beta]$. Retaining a fraction δ of $-\int_0^\beta\eta'^2$ before applying (30) and splitting $(qT+r)^2$ with a parameter κ, the coefficient of $\int_0^\beta\eta^2$ is

$$\left(\tfrac{1+\kappa}{1-\delta}-1\right)\tan^2\beta+1-\delta\left(\tfrac{\pi}{2\beta}\right)^2=-1.83\ldots<0 \qquad (\delta=\tfrac{1}{10},\ \kappa=1),$$

and the r^2 remainder, of coefficient $3.22\ldots$, is absorbed by the E_2 term of (24), which supplies $-\frac{1}{2}\int_{\pi/2}^{\pi-\beta}\eta'^2$: since $\eta(\pi/2)=0$ gives $\int_{\pi/2}^{\pi/2+\beta}\eta^2\le(\beta^2/2)\int\eta'^2$, the requirement is $3.22\cdot0.0420=0.135\le\frac{1}{2}$. So every ingredient of a Gårding inequality $\delta^2Q\le -c\|\eta\|_{L^2}^2$ is in place; Theorem 49 assembles it and Theorem 60 sharpens the constant to $\frac{2}{3}$.

The sharp value is $\frac{1}{2}c=0.7309566\ldots$ by Theorem 61. A P1 finite-element computation approaches it from above, as Rayleigh–Ritz must: $-0.735818, -0.733924, -0.733086, -0.732285, -0.731311, -0.7312$ at $m=32,\ldots,2048$.

13 Concavity, and the area identity

Theorem 49. *On the cell $\mathcal{C}$ of Σ's sign pattern, $\delta^2 Q \le 0$, with equality only at $\eta = 0$. Explicitly, writing $J_1 = \int_0^\beta \eta^2$, $J_2 = \int_\beta^{\pi/2} \eta^2$, $J_3 = \int_{\pi/2}^{\pi} \eta^2$,*

$$\tfrac{1}{2}\,\delta^2 Q[\eta] \;\le\; -1.6013\,J_1 - 0.3710\,J_2 + 0\cdot J_3 - 0.005\int_{\pi/2}^{\pi-\beta} \eta'^2. \tag{31}$$

Proof. On $[0,\beta)$ retain a fraction δ of $-\int_0^\beta \eta'^2$ and apply (30) with the remaining $-(1-\delta)\eta'^2$; completing the square in $p = \eta'(t)$ and discarding the resulting negative square,

$$-(1-\delta)p^2 - (qT+p)^2 + (r-p)^2 \;\le\; \frac{(qT+r)^2}{1-\delta} - q^2T^2 + r^2 \;\le\; A\,q^2T^2 + B\,r^2,$$

with $A = \frac{1+\kappa}{1-\delta} - 1$, $B = \frac{1+1/\kappa}{1-\delta} + 1$ from $(qT+r)^2 \le (1+\kappa)q^2T^2 + (1+1/\kappa)r^2$, and $T = \tan t \le \tan\beta$ on $[0,\beta)$. The three Poincaré constants for $\eta(0) = \eta(\pi/2) = 0$ are $P_1 = (\pi/2\beta)^2 = 29.4135$ on $[0,\beta]$, $P_2 = (\pi/2(\pi/2-\beta))^2 = 1.5035$ on $[\beta,\pi/2]$ and $P_3 = 1$ on $[\pi/2,\pi]$, each with a Dirichlet endpoint and a free one. Finally $-\big(\eta(t) + \eta'(t+\pi/2)\big)^2 \le -\lambda\,\eta'(t+\pi/2)^2 + \frac{\lambda}{1-\lambda}\eta(t)^2$ turns the E_2 term into $-\lambda\int_{\pi/2}^{\pi-\beta} \eta'^2 + \frac{\lambda}{1-\lambda}(J_1 + J_2)$, and $\eta(\pi/2) = 0$ gives $\int_{\pi/2}^{\pi/2+\beta} \eta^2 \le (\beta^2/2)\int_{\pi/2}^{\pi-\beta} \eta'^2$, so the $B\,r^2$ remainder is absorbed as soon as $\lambda \ge B\beta^2/2$. Collecting,

$$\text{coefficient of } J_1 = A\tan^2\beta - \delta P_1 + 1 + \tfrac{\lambda}{1-\lambda}, \qquad \text{of } J_2 = 1 - P_2 + \tfrac{\lambda}{1-\lambda},$$
$$\text{of } J_3 = 1 - P_3 = 0, \qquad \text{of } \int_{\pi/2}^{\pi-\beta} \eta'^2 = B\beta^2/2 - \lambda.$$

At $\delta = \frac{1}{10}$, $\kappa = 2$, $\lambda = 0.117$ these are -1.6013, -0.3710, 0 and -0.005, giving (31). For strictness, equality forces $\eta \equiv 0$ on $[0,\pi/2]$ and η the first Dirichlet–Neumann mode on $[\pi/2,\pi]$; but with $\eta \equiv 0$ on $[0,\pi/2]$ the E_2 term contributes $-\int_0^{\pi/2-\beta} \eta'(t+\pi/2)^2 < 0$. ☐

The bound $\frac{1}{2}\delta^2 Q \le -0.3710\|\eta\|^2_{L^2(0,\pi/2)}$ that (31) yields is about half of the sharp $-0.7309566\ldots$ for the full interval, the shortfall being the J_3 coefficient that (31) leaves at zero. Theorems 60 and 61 recover it.

Proposition 50. *$\sigma - \alpha_1 = \cos t\, x'(t)$ on each phase, with $x = (F-1)/\cos t$ as in (25). Hence the middle term of (20) is $\frac{1}{2}\int \cos^2 t\, x'(t)^2\,dt$, an energy of the intercept path.*

Proof. $\sigma - \alpha_1 = (F-1)\tan t + F'$ by (17) and (19), and $\cos^2 t\, x' = F'\cos t + (F-1)\sin t$ by (25); dividing by $\cos t$ gives the claim. Verified symbolically on SOL1, SOL5 and SOL6, residual 0 in each case, which also recovers the first and third closed forms of Proposition 30. ☐

Remark 51 (The area identity)**.** Since $u = 2\tan\beta$, Romik's area is

$$A_R^* = u^2 + 1 + \arctan(u/2) = 1 + 4\tan^2\beta + \beta.$$

Using Proposition 50 to make every integrand elementary, the six blocks of $|C_2|$ and the three blocks of V evaluate at 50 working digits to $|C_2| = 2.0133416126029788281721\ldots$, $V = 0.1841931970887689882517\ldots$, and

$$Q(\Sigma) - \big(1 + 4\tan^2\beta + \beta\big) = -2.7\cdot 10^{-51}.$$

A symbolic evaluation of the same integrals was attempted and did not terminate; the identity therefore remains HEURISTIC, now against a closed-form target and with a phase-by-phase decomposition.

Remark 52 (Status of Theorem 37)**.** Hypothesis (ii) is Theorems 41 and 45, proved from (RC). Hypothesis (i) splits: $\delta Q(\Sigma) = 0$ is Theorem 46, $\delta^2 Q \preceq 0$ is Theorem 49, and $Q(\Sigma) = A_R^*$ is Remark 51, still numerical. Hypothesis (iii), the separation $M < \frac{1}{2}$ for competitors, is untouched; across a family of low-frequency perturbations no competitor satisfying (RC) was found with $M \geq \frac{1}{2}$, but no implication is proved. So the conditional theorem now rests on one 51-digit numerical identity and one geometric hypothesis, rather than on four measured conditions. Optimality of Σ remains open.

14 Σ satisfies (RC), and $Q(\Sigma) = A_R^*$

Proposition 53. *Σ satisfies (RC). The surface-measure density is, block by block,*

$$H+H'' = \begin{cases} 0 & \theta \in [0,\beta) \text{ and } \theta \in (\pi-\beta,\pi], \\ \frac{1}{2} & \theta \in (\pi/2-\beta,\pi/2) \text{ and } \theta \in (\pi/2,\pi/2+\beta), \\ \frac{3K}{4}\cos\left(\frac{\theta}{2}+\frac{\pi}{8}\right) & \theta \in [\beta,\pi/2-\beta], \\ \frac{3K}{4}\sin\left(\frac{\theta-\pi/2}{2}+\frac{\pi}{8}\right) & \theta \in [\pi/2+\beta,\pi-\beta], \end{cases} \qquad K = f_1\sqrt{4-2\sqrt{2}},$$

plus the ceiling atom at $\theta = \pi/2$. Hence

$$\max(H+H'')_{\mathrm{ac}} = \tfrac{3K}{4}\cos\left(\tfrac{\beta}{2}+\tfrac{\pi}{8}\right) = 0.8388253494\ldots < 1,$$

and already the amplitude satisfies $\frac{3}{4}K = 0.9765387687\ldots < 1$, so (RC) holds as soon as

$$f_1^2 \;\leq\; \frac{16}{9\,(4-2\sqrt{2})} = 1.5174\ldots, \qquad \textit{against } f_1^2 = 1.4470\ldots \tag{32}$$

Proof. Each block of H is read off the three closed forms of Theorem 46. On the first phase $H = \cos\theta + \frac{1}{2}\sin\theta$, so $H + H'' = 0$; the shifted last phase is the same. On the last phase $H = (1-\frac{2}{3}a_1)\cos\theta + \frac{1}{2}\sin\theta + \frac{1}{2}$ and on the shifted first phase $H = (2a_1-1)\sin\theta + \frac{1}{2}\cos\theta + \frac{1}{2}$, each giving $H + H'' = \frac{1}{2}$. On the middle phase only the half-angle terms survive differentiation twice, contributing a factor $1 - \frac{1}{4}$, so $H + H'' = \frac{3}{4}(f_1\cos\frac{\theta}{2} + f_2\sin\frac{\theta}{2})$ and, on the shifted middle phase, $\frac{3}{4}(f_1\sin\frac{\theta}{2} - f_2\cos\frac{\theta}{2})$. Both have amplitude $\frac{3}{4}\sqrt{f_1^2+f_2^2} = \frac{3}{4}f_1\sqrt{4-2\sqrt{2}}$ by $f_2 = (1-\sqrt{2})f_1$, and phase $\pi/8$ because $-f_2/f_1 = \sqrt{2}-1 = \tan(\pi/8)$. On $[\beta,\pi/2-\beta]$ the argument $\frac{\theta}{2}+\frac{\pi}{8}$ runs over $[0.5375, 1.0333] \subset [0,\pi/2]$, where cosine decreases, so the maximum is at $\theta = \beta$. $\square$

Criterion (32) is strictly stronger than the separation criterion $f_1^2 < (2+\sqrt{2})/2 = 1.7071\ldots$ of Theorem 8: within the family in which f_1 is the free parameter, (RC) implies $M < \frac{1}{2}$.

Theorem 54. $Q(\Sigma) = A_R^*$.

Proof. Σ satisfies (RC) by Proposition 53, so $V(\Sigma) = |N(\Sigma)|$ by Theorems 41 and 45. The niches are disjoint by Theorem 8, and C_2 is ρ-invariant, so $|\Sigma| = |C_2| - |N| - |\rho N| = |C_2| - 2|N|$. Hence $Q(\Sigma) = |C_2| - 2V(\Sigma) = |C_2| - 2|N| = |\Sigma| = A_R^*$. $\square$

No integral is evaluated. Independently, with $A_R^* = 1+4\tan^2\beta+\beta$ and the integrands made elementary by Proposition 50, the two sides agree to $2.7\cdot 10^{-51}$ at 50 working digits, which is a consistency check on the whole chain rather than a proof of it.

Remark 55 (Regression against the one-corner problem)**.** The construction does *not* recover Gerver's sofa, and this should be stated plainly. Applying the same machinery to Gerver's rotation path:

- Gerver satisfies the one-corner injectivity condition everywhere: $\alpha_1 > 0$ and $\alpha_2 > 0$ on all of $[0, \pi/2]$, as expected.
- (RC) *fails* for Gerver's cap: $\max(H + H'')_{\mathrm{ac}} = 1.3960$ near $\theta = \pi$.
- Gerver's face-2 sweep is genuinely not injective in this decomposition: 33 002 of 158 802 tested pairs meet interior to both segments, with $\min(s_{22} - \alpha_2) = -0.0944$. The face-1 sweep is injective.
- Consistently, $V = 0.6426146$ against $|N| = |C| - |S| = 0.6406$ to 0.6412 at 481 and 721 constraints, i.e. $V > |N|$ as Proposition 28 requires.

So (RC) and the one-corner injectivity condition are independent: Σ satisfies the first and not the second, Gerver the second and not the first. By Remark 40 the difference has a classical reading: Σ's cap is a corridor facet plus a body that slides freely inside the unit disc, whereas part of Gerver's cap boundary is *flatter than the corridor is wide*, its curvature radius reaching 1.3986. The decomposition $N = W_2 \sqcup W_1^{\mathrm{out}}$ is therefore not the one the one-corner argument uses, and the fact that it fails on the solved case is a standing caution about the framework, recorded rather than explained.

Remark 56 (The separation hypothesis for competitors)**.** Hypothesis (iii) remains open. Over 250 random perturbations $H_\Sigma + \sum_{k=1}^{3} c_k \sin(2k\theta)$ with $|c_k| \le a/k^2$ for $a = 0.02, 0.05, 0.10$, every H satisfying (RC) had $M \le 0.3976$, against the required $M < \frac{1}{2}$; the (RC)-satisfying fraction fell from 250/250 to 34/250 as a grew. No counterexample and no implication.

15 Two negative findings

Remark 57 (The mirror decomposition does not rescue Gerver)**.** The decomposition of Proposition 20 has a mirror image under $t \mapsto \pi/2 - t$, $\mu \leftrightarrow \nu$: sweep the inner part of face 1 and the outer part of face 2, the latter reaching the floor at distance $\sigma_2 = c_y/\sin t$. For Gerver both give the same value, $0.6426144\ldots$, because Gerver's rotation path is itself symmetric under $t \mapsto \pi/2 - t$; both exceed $|N| \approx 0.641$. So the failure of Remark 55 is not an artefact of orientation.

The structural reason is visible in Proposition 53. Σ's cap has $H + H'' = 0$ on $[0, \beta)$ and $(\pi - \beta, \pi]$, i.e. *vertices*, because Σ's outer phases have constant contact arcs pinned at $(1, \frac{1}{2})$. Gerver's cap has no such vertices: there $H + H''$ reaches 1.3960 near $\theta = \pi$. The degeneracy that makes Σ hard for local analysis is exactly what makes (RC) hold for it.

Remark 58 (A false bound, retracted)**.** The following was attempted for hypothesis (iii) and is **false**. Solving $H'' + H = W$ with $H(0) = 1$ gives $H(\theta) = \cos\theta + H'(0)\sin\theta + \int_0^\theta \sin(\theta - s) W(s)\, ds$, and substituting into $c_y(t) = (H(t) - 1)\sin t + (H(t + \pi/2) - 1)\cos t$ yields

$$c_y(t) = H'(0) - \sin t - \cos t + \sin t \int_0^t \sin(t-s)W\,ds + \cos t \int_0^{t+\pi/2} \cos(t-s)W\,ds.$$

Both kernels are non-negative on their ranges, so "$W \le 1$" would give $c_y \le H'(0)$, and for a ρ-symmetric cap $H'(0) = \frac{1}{2} + \frac{1}{2}$(vertical right edge), apparently yielding $M \le \frac{1}{2}$.

The error is that W is *not* ≤ 1 as a measure: it carries the ceiling atom of mass $1.1670\ldots$ at $s = \pi/2$, and here that atom meets the kernel $\cos(t - \pi/2) = \sin t$, which does not vanish. The corrected bound is $M \le H'(0) + \frac{1}{2} \cdot 1.1670 = 1.083\ldots$, useless against the required $\frac{1}{2}$.

This does *not* affect Theorem 41: there the same atom lands at $u = t$, which is outside the integration range when $\tau < t$ and is the endpoint where the kernel $\sin(\tau - u)$ vanishes when $\tau = t$; and the face-1 atom needs $u = t - \pi/2 \le 0$. The placement of the atom relative to the kernel is what distinguishes the two arguments, and it was checked in both.

Hypothesis (iii) therefore remains open.

16 Relation to prior work

The one-corner problem was settled by Baek [1], who proved Gerver's sofa optimal. Our reading of that paper, for the purpose of separating what is his from what is not:

- The *arm lengths* f, g are his (Definition 6.2.1), defined through the contact points A and C; our α_1, α_2 are these shifted by the corridor width.
- His injectivity condition (Definition 6.1.2) has three parts. Part (1) requires the surface measure σ_K to be absolutely continuous on $[0, \pi/2)$ and on $(\pi/2, \pi]$, so an atom is permitted only at $\theta = \pi/2$; part (3) is $x'_K(t) \cdot u_t < 0$ and $x'_K(t) \cdot v_t > 0$ on $(0, \pi/2)$.
- Our (RC) shares part (1) verbatim and *adds* a bound on the density, while dropping part (3) entirely. No curvature bound appears in [1]; its only mentions of curvature are the standard identification of the surface-measure density with the radius of curvature, which we also use. The bound itself is classical, however: it is the sliding-ball condition of Remark 40, so (RC) is a known condition put to a new use rather than a new condition.
- His concavity argument attaches Mamikon regions to the overestimating region so that the total is linear in the cap; ours subtracts them from the cap. He works on a space of triples (K, B, D) of convex bodies; we work with the single support function H on $[0, \pi]$. Whether the two bookkeepings are equivalent we do not determine.
- The ambidextrous problem is not treated in [1], and we have found no upper bound for it in the literature.

The constants a_1, β, f_1 and the ODE families are Romik's [5]; the cubics of Section 6 are the standard ones and we claim no priority for them. This audit was made against [1] and the Kallus–Romik line [3] only, not against the wider literature.

17 The main theorem

Theorem 59. *Let $\mathcal{D}$ be the set of support functions H on $[0, \pi]$ of admissible caps such that*

(a) $H(0) = H(\pi/2) = 1$ (the gauge: horizontal translation, unit corridor)*;*

(b) (RC) σ_K *is absolutely continuous on* $[0, \pi/2)$ *and* $(\pi/2, \pi]$ *with density at most* 1*, the only atom being at* $\theta = \pi/2$*;*

(c) $c_y(t) = (H(t) - 1)\sin t + (H(t + \pi/2) - 1)\cos t \le \frac{1}{2}$ *for* $t \in [0, \pi/2]$ (note $\le$, not $<$: this gives $|U \cap \rho U| = 0$, which is all an area bound needs)*;*

(d) $\alpha_1 < 0$ *exactly on* $[0, \beta)$ *and* $\alpha_2 > 0$ *exactly on* $[0, \pi/2 - \beta)$*.*

Then $\mathcal{D}$ is convex, $\Sigma \in \mathcal{D}$, and

$$|T| \;\le\; A_R^*$$

for every ambidextrous moving sofa T whose cap data lies in $\mathcal{D}$.

Proof. Each of (a)–(d) is a family of affine inequalities or equalities in H, by Lemma 18 and (17)–(19), so $\mathcal{D}$ is convex. Σ satisfies (b) by Proposition 53, (c) with margin $\frac{1}{2} - M = 0.1121\ldots$ by Theorem 8, and (d) by definition of β.

By (b) and Theorems 41 and 45 the combined sweep is injective, so $V = |N|$ by Proposition 35, whose proof also supplies the covering and containment that Proposition 20 assumes. By (c) and Lemma 1, $U \subseteq \{y \le \frac{1}{2}\}$ and $\rho U \subseteq \{y \ge \frac{1}{2}\}$, so $|U \cap \rho U| = 0$; with C_2 ρ-invariant this gives $|T| \le |C_2| - |N| - |\rho N| = |C_2| - 2|N| = Q$. (Corollary 2 states the strict form $M < \frac{1}{2}$, which is what Σ satisfies; the null-set form used here is what the hypothesis $\le$ delivers.) By (d) and Theorem 49, $\delta^2 Q \le 0$ on $\mathcal{D}$; by Theorem 46, $\delta Q(\Sigma) = 0$. A concave function with a critical point on a convex set attains its maximum there, so $Q(H) \le Q(\Sigma)$, and $Q(\Sigma) = A_R^*$ by Theorem 54. Combining, $|T| \le Q(H) \le A_R^*$. □

The constant of Theorem 49 is far from the measured one, because the proof splits $[0, \pi/2]$ into $[0, \beta]$ and $[\beta, \pi/2]$ and applies a separate Poincaré inequality on each, and because it discards the E_2 term instead of spending it. Treating each half as a *single* weighted eigenvalue problem removes both losses.

Theorem 60 (Concavity, sharp order)**.** *On the cell $\mathcal{C}$, for every η with $\eta(0) = \eta(\pi/2) = 0$,*

$$\tfrac{1}{2}\,\delta^2 Q[\eta] \;\le\; -\tfrac{2}{3}\,\|\eta\|^2_{L^2(0,\pi)}. \tag{33}$$

Proof. Fix $\lambda \in (0, 1)$ and $\kappa \in (0, 1)$, and put $r = \lambda/(1 - \lambda)$ and $q = 1 + 1/\kappa$. Two elementary inequalities dispose of the two coupling terms of (24). Since $(1 + r)(1 - \lambda) = 1$,

$$(1{+}r)a^2{+}2ab{+}(1{-}\lambda)b^2 = \left(\sqrt{1+r}\,a{+}\sqrt{1-\lambda}\,b\right)^2 \;\ge\; 0, \qquad \text{that is} \qquad -(a{+}b)^2 \;\le\; -\lambda\, b^2{+}r\,a^2,$$

applied with $a = \eta(t)$, $b = \eta'(t + \pi/2)$ on $E_2 = [0, \pi/2 - \beta)$; it buys gradient weight λ on $[\pi/2, \pi - \beta]$ at the price of mass r on $[0, \pi/2 - \beta)$. And $(x - y)^2 \le (1 + 1/\kappa)x^2 + (1 + \kappa)y^2$ with $x = \eta(t + \pi/2)$, $y = \eta'(t)$ on $E_1 = [0, \beta)$ pays mass q on $[\pi/2, \pi/2 + \beta]$ and gradient weight $1 + \kappa$ on $[0, \beta)$. Substituting $s = t + \pi/2$ in both, the two halves decouple:

$$\tfrac{1}{2}\delta^2 Q[\eta] \;\le\; \int_0^{\pi/2} \left(m_L \eta^2 - w_L \eta'^2\right) + \int_{\pi/2}^{\pi} \left(m_R \eta^2 - w_R \eta'^2\right) \;=:\; B[\eta],$$

$$w_L = 1-\kappa \text{ on } [0,\beta),\ 2 \text{ after}, \qquad m_L = 2+r \text{ on } [0,\tfrac{\pi}{2}-\beta),\ 2 \text{ after},$$
$$w_R = 1+\lambda \text{ on } [\tfrac{\pi}{2},\pi-\beta),\ 1 \text{ after}, \quad m_R = 1+q \text{ on } [\tfrac{\pi}{2},\tfrac{\pi}{2}+\beta),\ 1 \text{ after}.$$

Each half is a Rayleigh quotient, so $B[\eta] \le -\min(c_L, c_R)\,\|\eta\|^2_{L^2(0,\pi)}$, where c_L and c_R are the first eigenvalues of

$$-(w_L\eta')' - m_L\eta = c\,\eta \ \text{ on } (0,\tfrac{\pi}{2}), \quad \eta(0)=\eta(\tfrac{\pi}{2})=0; \qquad -(w_R\eta')' - m_R\eta = c\,\eta \ \text{ on } (\tfrac{\pi}{2},\pi), \quad \eta(\tfrac{\pi}{2})=0,\ \eta'(\pi)=0.$$

The Dirichlet conditions at 0 and $\pi/2$ are the gauge, not a restriction, and nothing pins $\eta(\pi)$.

Both coefficients are piecewise constant with three pieces, so the solution of the initial value problem $\eta = 0$, $w\eta' = 1$ at the left endpoint is a product of three transfer matrices acting on $(\eta,\, w\eta')^\top$,

$$T(\ell, w, m; c) = \begin{pmatrix} \cos k\ell & (wk)^{-1}\sin k\ell \\ -wk\sin k\ell & \cos k\ell \end{pmatrix}, \qquad k = \sqrt{(m+c)/w}$$

(hyperbolic when $m+c<0$). Write $\Phi_L(c)$ for the resulting $\eta(\pi/2)$ and $\Phi_R(c)$ for $(w\eta')(\pi)$. Its zeros are exactly the eigenvalues, and by Sturm oscillation $\Phi > 0$ on $(-\infty, c_1)$ — so a single sign evaluation bounds c_1 from *below*, which is the direction a certificate needs and the one a Rayleigh–Ritz or finite-element computation cannot supply.

Positivity alone does not locate c, because $\Phi > 0$ again on (c_2, c_3); one also needs $c < c_2$. That comes from min–max, crudely. Since $\int w\eta'^2 \ge w_{\min}\int \eta'^2$ and $-\int m\eta^2 \ge -m_{\max}\int\eta^2$, the form dominates the constant-coefficient one, so $c_k \ge w_{\min}\lambda_k - m_{\max}$ with λ_k the eigenvalues on an interval of length $\pi/2$, namely $(2k)^2$ for Dirichlet–Dirichlet and $(2k-1)^2$ for Dirichlet–Neumann. Hence

$$c_2^L \ \ge\ \tfrac{3}{4}\cdot 16 - \tfrac{63}{13} = 7.153\ldots, \qquad c_2^R \ \ge\ 1\cdot 9 - 6 = 3,$$

both above $\frac{2}{3}$. (At $k=1$ the same bound gives -1.85 and -5, useless; the second eigenvalue is far enough away to be reached crudely while the first is not, which is exactly why the transfer matrix is needed.)

It remains to evaluate two signs. At $\kappa = \frac{1}{4}$, $\lambda = \frac{37}{50}$ and $c = \frac{2}{3}$ every weight and mass is rational — $w_L = \frac{3}{4}, 2, 2$ and $m_L = \frac{63}{13}, \frac{63}{13}, 2$; $w_R = \frac{87}{50}, \frac{87}{50}, 1$ and $m_R = 6, 1, 1$ — so each $k^2 = (m+c)/w$ is exact, and the only inexact inputs are the three lengths β, $\frac{\pi}{2}-2\beta$, β and the trigonometric functions of $k\ell$. In ball arithmetic at 300 bits, with $\beta = \arctan(u/2)$ enclosed from $u^3 + 3u = 2$ by exact rational bisection,

$$\Phi_L(\tfrac{2}{3}) \in 0.0046739486452026048\ldots \pm 6.4\cdot 10^{-73}, \qquad \Phi_R(\tfrac{2}{3}) \in 0.0026447861413929500\ldots \pm 8.3\cdot 10^{-73},$$

both certified strictly positive. Hence $c_L, c_R > \frac{2}{3}$, which is (33). □

Optimising the two parameters gives $\min(c_L, c_R) = 0.674944$ at $(\kappa,\lambda) = (0.26, 0.74)$. Since $\|\eta\|_{L^2(0,\pi/2)} \le \|\eta\|_{L^2(0,\pi)}$, (33) also improves the $[0,\pi/2]$ constant of Theorem 49 from 0.371 to $\frac{2}{3}$.

The splitting into two scalar problems is itself avoidable, and avoiding it gives the sharp constant.

Theorem 61 (The sharp constant)**.** *The best constant in* (33) *is the first eigenvalue* c^* *of an explicit* 2×2 *Sturm–Liouville system, and*

$$\tfrac{1}{2}\,\delta^2 Q[\eta] \ \le\ -\tfrac{73}{100}\,\|\eta\|^2_{L^2(0,\pi)}, \qquad c^* = 0.7309566\ldots \tag{34}$$

Proof. The two halves of $[0,\pi]$ are the same interval re-indexed. Put $p(t) = \eta(t)$ and $q(t) = \eta(t+\pi/2)$ for $t \in [0,\pi/2]$; then (24) reads $\frac{1}{2}\delta^2 Q = \int_0^{\pi/2} L$ with

$$L = 2p^2 - 2p'^2 + q^2 - q'^2 - \mathbf{1}_{E_2}(p+q')^2 + \mathbf{1}_{E_1}(q-p')^2,$$

and every boundary condition is forced: $p(0) = p(\pi/2) = 0$ is the gauge, $q(0) = \eta(\pi/2) = 0$ is the same gauge seen from the other half, and $q(\pi/2) = \eta(\pi)$ is free. Nothing has been discarded, so the first eigenvalue of this system is the best constant. Writing $M := -L$ as $ap'^2 + bq'^2 + gp^2 + dq^2 + 2e\,pq' + 2f\,qp'$,

piece	a	b	g	d	e	f
$[0,\beta)$	1	2	-1	-2	1	1
$[\beta, \frac{\pi}{2}-\beta)$	2	2	-1	-1	1	0
$[\frac{\pi}{2}-\beta, \frac{\pi}{2})$	2	1	-2	-1	0	0

On $[0,\beta)$ — exactly where the obstruction sits — $e = f = 1$ and $2pq' + 2qp' = 2(pq)'$ is a total derivative: the E_1 obstruction and the E_2 resource cancel there up to the single boundary value $2p(\beta)q(\beta)$. The coupling is not pointwise, which is why every pointwise splitting loses and this one does not.

Eliminating p', q' in favour of the momenta $P = ap' + fq$ and $Q = bq' + ep$, the Euler–Lagrange system is first order in (p, P, q, Q) with piecewise constant matrix

$$A(c) = \begin{pmatrix} 0 & 1/a & -f/a & 0 \\ g - c - e^2/b & 0 & 0 & e/b \\ -e/b & 0 & 0 & 1/b \\ 0 & f/a & d - c - f^2/a & 0 \end{pmatrix},$$

so the transfer across a piece of length ℓ is $\exp\big(A(c)\ell\big)$. The left conditions $p(0) = q(0) = 0$ leave a two-dimensional solution space; propagating a basis through the three pieces and imposing $p(\pi/2) = 0$ and $Q(\pi/2) = 0$ gives a 2×2 determinant $\Phi(c)$ whose zeros are exactly the eigenvalues. A sign change brackets $c^* \in (0.7309, 0.7310)$.

For the certificate the oscillation argument is unavailable — for a system the count is a Maslov index, not a zero count — and it is not needed. Two facts compose: $c^* \geq \frac{2}{3}$ by Theorem 60, and Φ has no zero on $[\frac{2}{3}, \frac{73}{100}]$. The latter is checked by covering that interval with 60 balls and verifying in 400-bit ball arithmetic that no enclosure of Φ contains 0, with β enclosed from $u^3 + 3u = 2$ by exact rational bisection. Hence no eigenvalue lies at or below $\frac{73}{100}$. ☐

The certified $\frac{73}{100}$ is 99.9% of c^*; the decoupled $\frac{2}{3}$ is 91.2%. Both are lower bounds, and no step in either is floating point. Nothing in the argument stops at $\frac{73}{100}$: certifying a target c needs the enclosures to separate from 0 on every subinterval of $[\frac{2}{3}, c]$, and those must shrink as c approaches the root, so the number of balls grows — $60, 60, 800, 12000$ for targets $0.73, 0.7305, 0.7309, 0.73095$. The last is 99.9991% of c^*. We quote $\frac{73}{100}$ because a reader can check it with 60 balls; the limit is arithmetic cost, not a gap.

Remark 62 (The $[\pi/2, \pi]$ eigenvalue in closed form)**.** When only the weight is raised and the mass is left at 1 – that is, $w = 1 + \lambda_J$ on $[\pi/2, \pi - \beta]$ and 1 on $[\pi - \beta, \pi]$ – the transfer product collapses to a transcendental equation for the first eigenvalue $\Lambda(\lambda_J)$ of $-(w\eta')' = \Lambda\eta$ with $\eta(\pi/2) = 0$, $\eta'(\pi) = 0$:

$$\sqrt{w_1}\,\cot\big(L_1\sqrt{\Lambda/w_1}\big) = \tan\big(L_2\sqrt{\Lambda}\big), \qquad w_1 = 1 + \lambda_J, \quad L_1 = \tfrac{\pi}{2} - \beta, \quad L_2 = \beta. \tag{35}$$

At $\lambda_J = 0$ this reads $\cot(L_1\sqrt{\Lambda}) = \cot(\pi/2 - L_2\sqrt{\Lambda})$, so $\sqrt{\Lambda}\,(L_1+L_2) = \pi/2$; since $L_1+L_2 = \pi/2$ *exactly*, $\Lambda(0) = 1$ exactly. That is the marginal case $P_3 = 1$ of Theorem 49, and it explains why no gain is available on $[\pi/2, \pi]$ until the weight is raised. For $\lambda_J > 0$ put $G(w_1) = \sqrt{w_1}\cot(L_1/\sqrt{w_1}) - \tan\beta$. Then $G(1) = 0$, and G increases: the prefactor increases, while $L_1/\sqrt{w_1}$ decreases and cot decreases on $(0, \pi)$. So $G > 0$ for $w_1 > 1$. The left side of (35) decreases in Λ and the right side increases, so the root moves right and $\Lambda(\lambda_J) > 1$ for every $\lambda_J > 0$:

λ_J	0	0.05	0.10	0.16483	0.25	0.35
Λ	1	1.049467	1.098883	1.162878	1.246819	1.345183

Theorem 60 runs this mechanism on both halves at once, raising the mass as well as the weight.

Because Q is exactly quadratic on the cell and $\delta Q(\Sigma) = 0$, the same ingredients give more than a bound.

Theorem 63 (Stability)**.** *For every ambidextrous moving sofa T whose cap data $H = H_\Sigma + \eta$ lies in $\mathcal{D}$,*

$$|T| \;\le\; A_R^* - \tfrac{73}{100}\,\|\eta\|^2_{L^2(0,\pi)}. \tag{36}$$

Proof. On the cell $\mathcal{C}$ the functional Q is a quadratic polynomial in H, so $Q(H_\Sigma + \eta) = Q(\Sigma) + \delta Q(\Sigma)[\eta] + \frac{1}{2}\delta^2 Q[\eta]$ exactly. The first variation vanishes by Theorem 46 and $Q(\Sigma) = A_R^*$ by Theorem 54, while Theorem 61 bounds the second variation. Combining with $|T| \le Q(H)$ from Theorem 59 gives (36). □

Corollary 64 (Uniqueness)**.** *Σ is the unique ambidextrous moving sofa of area A_R^* whose cap data lies in $\mathcal{D}$; every other one has strictly smaller area, by the amount* (36)*.*

Proof. Equality in (36) forces $\|\eta\|_{L^2(0,\pi/2)} = 0$, and then $\eta \equiv 0$ on $[0, \pi/2]$; Theorem 49 shows $\delta^2 Q < 0$ strictly unless $\eta \equiv 0$ throughout, since with $\eta \equiv 0$ on $[0, \pi/2]$ the E_2 term of (24) contributes $-\int_0^{\pi/2-\beta} \eta'(t+\pi/2)^2 dt < 0$ unless η is constant on $[\pi/2, \pi]$, and $\eta(\pi/2) = 0$ makes that constant zero. □

The domain $\mathcal{D}$ is cut out in part by (RC), and (RC) enters at exactly one point: it forces the flux excess $E := V - |N|$ to vanish, so that Q is tight. Separation and Reynolds give, with no curvature hypothesis,

$$|T| \;=\; |C_2| - 2|N| \;=\; Q + 2E, \qquad E \ge 0. \tag{37}$$

It is natural to ask for a concave majorant of E, which would replace $\{E = 0\}$ by something larger. There is none.

Proposition 65 (No concave majorant)**.** *Let $\mathcal{D}'$ be convex with Σ in its algebraic interior, and let $\widetilde{E} : \mathcal{D}' \to \mathbb{R}$ be concave with $\widetilde{E} \ge 0$ and $\widetilde{E}(\Sigma) = 0$. Then $\widetilde{E} \equiv 0$, and hence $E \equiv 0$ on $\mathcal{D}'$.*

Proof. Let $a, b \in \mathcal{D}'$ with $\Sigma = \frac{1}{2}(a+b)$. Concavity gives $0 = \widetilde{E}(\Sigma) \ge \frac{1}{2}\widetilde{E}(a) + \frac{1}{2}\widetilde{E}(b) \ge 0$, so $\widetilde{E}(a) = \widetilde{E}(b) = 0$. As Σ is interior, every point of $\mathcal{D}'$ arises this way. □

So any architecture with a concave, tight majorant is confined to the injectivity locus. The hypothesis to weaken is concavity of $\widetilde{E}$: what the argument needs is that $Q + 2\widetilde{E}$ be concave, and the strict concavity of Q leaves room.

Theorem 66 (Excess budget)**.** *Let $\mathcal{D}'$ be convex with $\Sigma \in \mathcal{D}'$ and $\theta \in [0,1]$. Suppose $M < \frac{1}{2}$ on $\mathcal{D}'$, that $\frac{1}{2}\delta^2 Q \le -c\|\eta\|^2_{L^2(0,\pi)}$ there, and that*

$$E(H) \;\le\; \tfrac{1}{2}\,\theta\, c\,\|H - H_\Sigma\|^2_{L^2(0,\pi)} \qquad \text{on } \mathcal{D}'. \tag{38}$$

Then $|T| \le A^*_R - (1-\theta)\,c\,\|H - H_\Sigma\|^2_{L^2(0,\pi)}$ for every ambidextrous moving sofa T with cap data in $\mathcal{D}'$.

Proof. Concavity of Q with $\delta Q(\Sigma) = 0$ gives $Q(H) \le A^*_R - c\|\eta\|^2$. Adding (37) and (38), $|T| = Q + 2E \le A^*_R - c\|\eta\|^2 + \theta c\|\eta\|^2$. □

At $\theta = 0$ this is Theorem 63. For $\theta > 0$ it admits $E > 0$, so Proposition 65 is evaded, and the room available is proportional to c: the constant $c^* = 0.7309566$ of Theorem 61 is what makes (38) usable. Measured along $H_\Sigma + \varepsilon\phi$ with ϕ the bump at $\theta_0 = 1$ of half-width 0.45, the least admissible θ is

ε	0.0141	0.0200	0.0283	0.0400	0.0566	0.0800
θ	0.100	0.301	0.683	1.276	2.090	3.106

so $\theta < 1$ holds out to $\varepsilon \approx 0.031$, against $\varepsilon_c = 0.006492$ for (RC): the admissible radius grows by a factor of about 4.8 in that direction. This is a measurement, not a proof.

Proposition 67 (Beyond Σ's own cell)**.** *Q is C^1 and piecewise quadratic, so if every cell met by the segment $[H_\Sigma, H]$ has $\delta^2 Q \le 0$, then Q is concave along that segment and $Q(H) \le Q(\Sigma) + \delta Q(\Sigma)[H - H_\Sigma] = A^*_R$. Convexity of the union of those cells is not needed. In particular this holds whenever the segment meets only cells whose sign sets are the anchored intervals $E_1 = [0,\tau_1)$, $E_2 = [0,\tau_2)$ with $\tau_1 \le \tau_2$: across $(\tau_1,\tau_2) = (0,\frac{\pi}{2}), (0.1,0.2), (0.3,0.5), (0.5,0.9), (0.8,1.2), (1.0,1.3), (1.2,1.4), (1.4,1$ the values of $\sup \frac{1}{2}\delta^2 Q/\|\eta\|^2_{L^2(0,\pi)}$ at $m = 128$ are $-0.8751, -0.1404, -0.3602, -0.5397, -0.4497, -0.3431, -0.2285,$ all negative, with Σ's own cell giving -0.7331 (at $m = 128$; the converged value is -0.7309566 by Theorem 61).*

Since Σ sits at $(\beta, \pi/2 - \beta)$ with $\beta < \pi/2 - \beta$, it lies in that family, and the class of competitors covered is strictly larger than $\mathcal{D}$ and is checkable from the sign pattern alone. (The restriction to *anchored* intervals is essential: Remark 26 exhibits $E_1 = E_2 = [0.4, 1.2]$, unanchored, with $\sup \frac{1}{2}\delta^2 Q/\|\eta\|^2_{L^2(0,\pi)} = +0.4123$.)

The constant in (36) is 99.9% of the sharp $c^* = 0.7309566\ldots$. What closed the earlier factor of 4.8 was giving up the pointwise splitting in two stages: Theorem 49 uses three separate Poincaré constants, of which $P_3 = 1$ is marginal and $P_2 = 1.5035$ nearly so; Theorem 60 replaces all three by two weighted eigenvalues that see the whole of each half at once; and Theorem 61 replaces those two by one system that sees both halves at once, losing nothing.

A correction this forced. Earlier drafts quoted the sharp constant as 0.7323, the P1 value at $m = 256$. Rayleigh–Ritz gives an upper bound on an eigenvalue and that sequence had not converged; it decreases through $0.731311, 0.731247, 0.731073$ at $m = 512, 1024, 2048$ towards $c^* = 0.7309566$. Every certificate in this note is a lower bound and every finite-element number an upper bound, so nothing crossed and no conclusion moves; the target was mis-stated by 0.2%.

Proposition 68 (H^1 stability)**.** *The second variation controls the derivative as well: measured against correctly assembled mass and stiffness matrices, the largest c_0 with $\frac{1}{2}\delta^2 Q[\eta] \le -c_0\|\eta\|^2_{L^2(0,\pi)} - c_1\|\eta'\|^2_{L^2(0,\pi)}$ is*

c_1	0	0.1	0.3	0.5	0.7	1.0
c_0	0.7323	0.6114	0.3571	0.0760	−0.2560	−1.0000

at $m = 256$, each converging in m. So $\frac{1}{2}\delta^2 Q \le -0.357\|\eta\|_{L^2}^2 - 0.3\|\eta'\|_{L^2}^2$, and the frontier is close to $c_0 = 0.732 - 1.31\,c_1$, vanishing near $c_1 = 0.56$.

Two simplifications are worth recording. First, the σ term has no singularity: since $-2\int \eta\eta' \tan t = -\int(\eta^2)' \tan t = \int \eta^2 \sec^2 t$ when $\eta(0) = \eta(\pi/2) = 0$, and $\sec^2 t - \tan^2 t = 1$,

$$-\int_0^{\pi/2} \big(\eta \tan t + \eta'\big)^2 dt = \int_0^{\pi/2} \big(\eta^2 - \eta'^2\big) dt,$$

checked to 10^{-12} on five test functions, so (24) may be written with bounded coefficients throughout. Second, the sharp L^2 constant is 0.7309566, not the 0.7285 of an earlier draft; that figure came from a mass matrix that gave the endpoint hat at $\theta = \pi$ the interior weight $2h/3$ instead of $h/3$.

Remark 69 (The size of $\mathcal{D}$, and what Theorem 59 is not)**.** It is not a proof that Σ is optimal, and the reason is quantitative.

Measuring the radius of $\mathcal{D}$ along $H_\Sigma + \varepsilon \sin(2k\theta)$ gives $\varepsilon_k = 0.0864,\ 0.0117,\ 0.0047,\ 0.0036,\ 0.0021$ at $k = 2, 4, 6, 8, 10$, so $\varepsilon_k k^2$ stays between 0.17 and 0.35: the radius scales like k^{-2}. $\mathcal{D}$ therefore contains a ball in the C^2 norm and in no weaker one, which is exactly what a bound on H'' should give. It is a genuine ball rather than a sliver — of 60 random eight-mode perturbations with coefficients of size $0.02/k^2$, 51 land inside — but every member is a small C^2-perturbation of Σ. **Theorem 59 is thus a C^2-local statement**, with explicit convex hypotheses in place of a neighbourhood of unspecified size.

In every one of those probes the binding constraint was (b); condition (c) never became active. That is a statement about the probe and not about the geometry: (c) is *independent* of (b). Adding $\delta \sin(\theta - \pi/2)$ on $[\pi/2, \pi]$ and 0 on $[0, \pi/2]$ leaves $H + H''$ identically unchanged, so (b) and the gauge hold exactly, while the ceiling facet grows by δ and c_y grows by $\delta \sin t \cos t$; at $\delta = 0.2243\ldots$ the maximum M crosses $\frac{1}{2}$. So (c) cannot be derived and must remain a hypothesis.

$\mathcal{D}$ does not reach Gerver's cap, and by Remark 70 injectivity genuinely fails once the density exceeds 1. The failure is sharp in the condition but gentle in magnitude: comparing the flux $\frac{1}{2}\int(\alpha_2^+)^2$ with the area actually swept, the excess is $5.4 \cdot 10^{-7}$ at density 1.047 and $7.4 \cdot 10^{-5}$ at 1.267, and Gerver's own excess is about 0.25%. Widening $\mathcal{D}$ therefore does not need a weaker constant; it needs an upper bound on that excess which is *concave* in H, so that subtracting it preserves the concavity of Q. Truncating the sweep at $\inf_{t'} s_{22}(t, t')$ fails because that infimum is concave in H and $\frac{1}{2}\lambda^2$ then loses the convexity the Mamikon step supplies; the first-capture decomposition is exact but its parameter domain stops being of the form $0 \le s \le \alpha_2$, so its area is no longer quadratic in H. So the honest reading is that the one-corner architecture transfers to the ambidextrous problem on an explicit convex C^2-neighbourhood of Σ, and that enlarging that neighbourhood is where the remaining difficulty sits.

Remark 70 ((RC) is sharp)**.** The constant 1 in the forcing $R = 1 - (G + G'')$ of Theorem 41 is the corridor width: it enters through $c(t) = (F - 1)\mu_t + (G - 1)\nu_t$, so $\langle c(t), \nu_{t'}\rangle = n + G - 1$ and $n'' = -n + 1 - (G + G'')$. The argument therefore cannot tolerate a density above 1. Nor can the conclusion: perturbing Σ so that the maximum density passes through 1, the face-2 sweep acquires self-intersections exactly there. At densities $0.838, 0.921, 0.976$ there are none; at $1.004, 1.047, 1.119$ there are $160, 1510, 3086$ among $\binom{301}{2}$-many tested pairs, and a second mode gives 0 at 0.976 against 200 and 526 at 1.045 and 1.114. So (b) is sharp as stated.

18 Formalisation

Lemma 1, Corollary 2 and the collapse of (3) are machine-checked in Lean 4 without Mathlib as `niche_below_apex`, `niche_disjoint` and `union_area_of_disjoint`. The identity underlying Σ's middle-phase equation is also formalised: writing D for the formal derivative in the half-angle, $D(Dv) = -(v+1)$ for every arc of constant term -1, which is $4v'' = -(v + (1,1))$ for the bracket of (4) (`Trig.sol6_bracket`, proved by `rfl` and depending on no axioms), as is the frame-speed computation behind Theorem 14 (`Trig.sol1_speed_mu`) and the covering criterion of Theorem 11 (`strip_covers_iff`). `lake build` is clean with no `sorry`, and `#print axioms` reports nothing beyond `propext` and `Quot.sound`. Trigonometric quantities are carried as formal symbols with the sign hypotheses $\sin t \geq 0$, $\cos t \geq 0$, which is all Lemma 1 uses.

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